

HPS Gaussian-Process Resonance Search

Selected Sections 2–6: Data, Method, Validation, and Results

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September 2, 2026

Submitted by Emrys Peets. This excerpt comes from a collaborative Heavy Photon Search analysis note; all three note authors are listed above. HPS searches for a narrow resonance in the reconstructed e^+e^- mass spectrum. Sections 2–5 describe the data sets, detector inputs, Gaussian-process search method, and background-model validation. Section 6 presents selected observed upper limits and local fixed-mass diagnostics, including the strongest fluctuation in the common three-campaign mass range.

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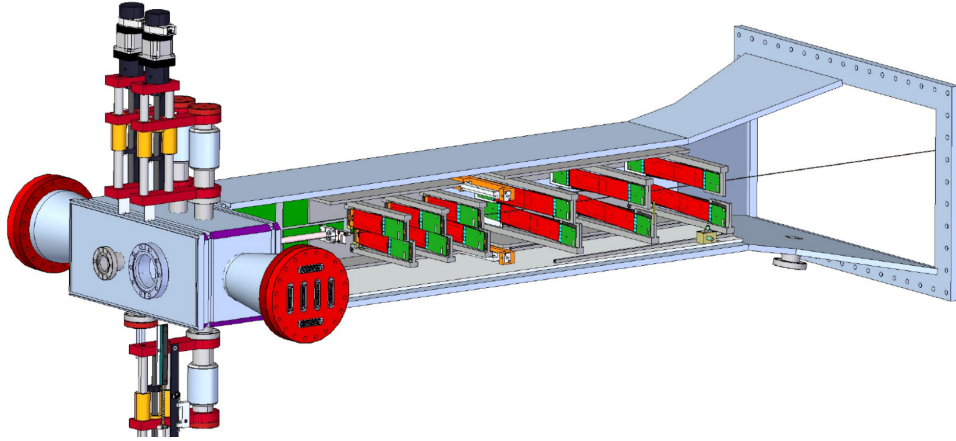
2 HPS apparatus and data sets

2.1 Detector configurations for the prompt search

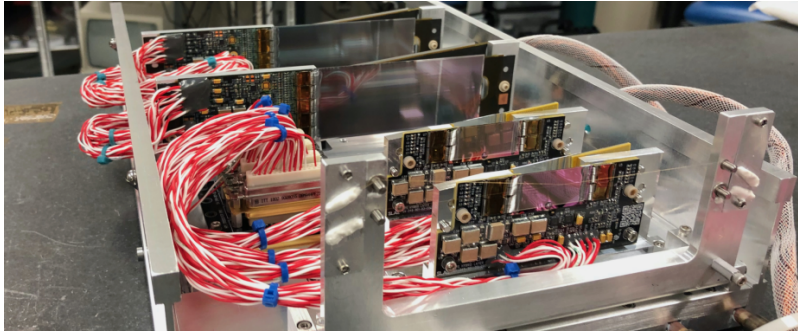
HPS is a forward spectrometer optimized for narrow e^+e^- resonances and displaced vertices in the high-rate environment immediately downstream of a thin tungsten target. Its prompt $A' \rightarrow e^+e^-$ sensitivity is set primarily by three subsystems: the Silicon Vertex Tracker (SVT), which determines the mass and vertex resolution; the electromagnetic calorimeter (ECal), which provides the trigger and track-cluster matching; and the beamline/target geometry, which defines the forward acceptance and the occupancy environment [1, 2].

The 2015 and 2016 engineering runs used the baseline six-station SVT. In that configuration, six axial/stereo tracking stations were distributed between $z = 10$ cm to 90 cm downstream of the target, with the first active silicon only about 1.5 mm from the beam plane [3, 1]. The ECal geometry used in the engineering runs was already close to the final HPS layout: two PbWO_4 crystal halves with the highest-rate crystals nearest the beam removed, leaving the beam to pass between the two halves [3, 1].

Before the first physics run in 2019, HPS upgraded the front tracker. The 2019 and 2021 configurations added a seventh double layer at $z = 5$ cm and used thin, slim-edge sensors with short split triplets [1, 4]. Moving the first precision measurements closer to the beam reduced material and occupancy while improving acceptance for low-mass, small-opening-angle pairs. The ECal crystal layout was unchanged, but a positron hodoscope was installed before the 2019 run to support a single-arm trigger for events near the ECal beam gap [1]. Figure 1 compares the tracker layouts used in the engineering and physics runs.



(a) Baseline six-layer SVT used in the 2015 and 2016 engineering runs [3].



(b) Front-layer upgrade used in the 2019 and 2021 physics runs, including the added seventh double layer and slim-edge front sensors [1, 4].

Figure 1: SVT configurations used in the prompt-resonance analyses. The 2015/2016 baseline tracker and the 2019/2021 tracker differ in both layer count and the design of the innermost stations. The ECal crystal layout was unchanged; the physics-run configuration added a positron hodoscope.

2.2 Data-taking campaigns and analysis scope

This note contains two related analysis states. The accepted v4.2 combined result uses the full 2015 and 2016 engineering-run samples together with a distributed 10% sample from the 2021 physics run. The later v4.9.5 pass examines that 2021 sample alone with a different lower support edge. The engineering data share the baseline tracker used in the published prompt searches; the 2021 data were recorded with the upgraded tracker and positron hodoscope. Although the 2019 run used the same upgraded detector, no 2019 data enter either analysis.

Table 1 summarizes the run periods, analyzed samples, search intervals, and GP fit supports. A mass hypothesis is tested only inside the search interval, whereas the GP is trained on the wider support after removing the local $2.25 \sigma_m$ exclusion.

Run period	Beam energy	Sample	Search interval	v4.2 GP fit support
2015 engineering run	1.056 GeV	1170 nb ⁻¹	19 MeV to 90 MeV	14 MeV to 135 MeV
2016 engineering run	2.3 GeV	10 608 nb ⁻¹	39 MeV to 180 MeV	30 MeV to 210 MeV
2021 physics run	3.74 GeV	Native 10% sample from a ~ 160 pb ⁻¹ parent run	50 MeV to 250 MeV	40 MeV to 300 MeV

Table 1: Data sets and mass domains in the accepted v4.2 three-campaign configuration. The 2015 and 2016 inputs are the full engineering-run samples; the 2021 input is the distributed native 10% sample. The wider GP supports provide training sidebands beyond both ends of each search interval. The 2016 lower edge is 30 MeV because a 35 MeV edge leaves no rebinned training-bin center below the exclusion at the 39 MeV search endpoint. The corresponding published prompt searches covered 19 MeV to 81 MeV in 2015 and 39 MeV to 179 MeV in 2016 [5, 2].

The later v4.9.5 study changes only the lower support edge for a 2021-only extraction: it uses 36–300 MeV while retaining the 50–250 MeV search interval. The accepted v4.2 three-campaign configuration in Table 1 continues to use 40–300 MeV for 2021.

Integrated luminosity and analyzed fraction do not enter the GP interpolation. They enter later, with the target, acceptance, efficiency, and radiative fraction, when an extracted prompt yield is converted to an equivalent ϵ^2 limit.

The 2015 and 2016 samples are fully unblinded. The 2021 input is distributed across the run rather than drawn from a contiguous block, so it samples varied run conditions while leaving most of the exposure unopened. The ROOT file does not contain enough exposure metadata to infer an integrated luminosity; “10%” describes how the sample was constructed. The three-campaign result combines the full 2015 and 2016 samples with this 2021 sample and is not the final result for the 2021 run.

Figure 2 compares the unit-normalized invariant-mass shapes. Figure 3 shows the same spectra in events per MeV and marks the v4.2 fit supports and search intervals.

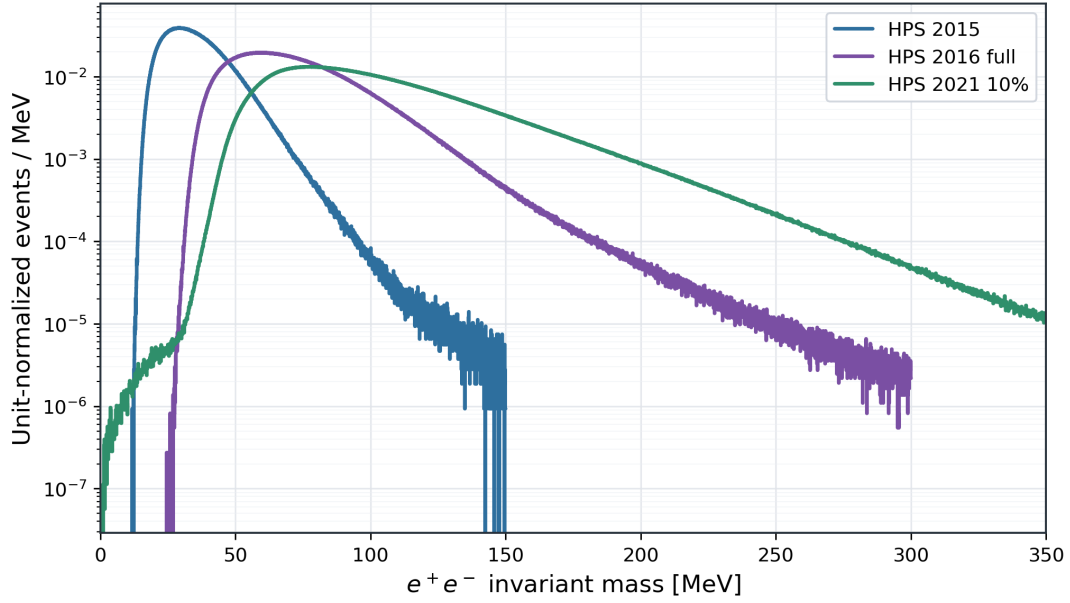


Figure 2: Unit-normalized invariant-mass shapes for the full 2015 and 2016 samples and the native 2021 10% sample. Each histogram is divided by its event count and bin width, so the comparison shows spectral shape rather than sample size.

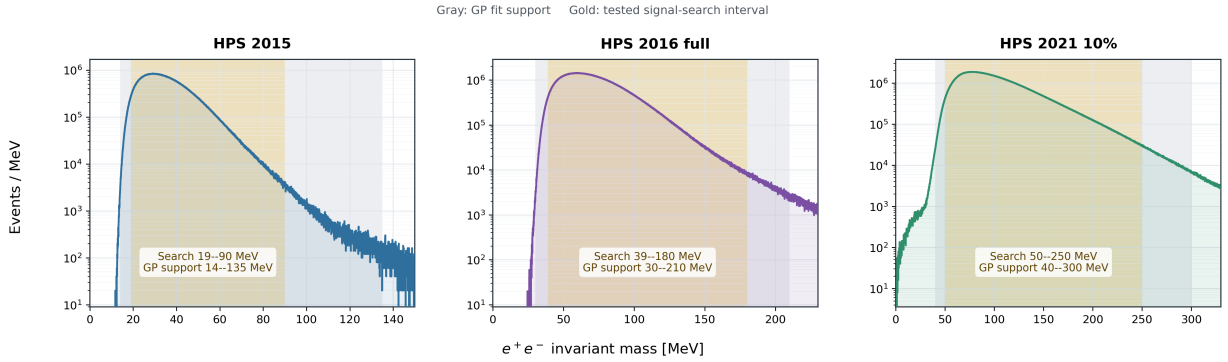


Figure 3: Invariant-mass distributions for the three inputs in the accepted v4.2 three-campaign configuration. Gray shading marks the GP fit support and gold shading marks the narrower interval in which signal hypotheses are tested. The mass-dependent $2.25\sigma_m$ exclusion is removed from the GP support separately at every tested mass.

3 Event selection, mass resolution, and radiative fraction

Each HPS input is an invariant-mass spectrum formed after a run-specific event selection. The selection fixes the composition of the prompt e^+e^- sample, while the detector response sets the signal width and the radiative fraction sets its normalization. This section describes those three ingredients together.

3.1 Prompt event selection

Across the three HPS campaigns, the prompt selection follows the same experimental logic: oppositely charged tracks matched to ECal clusters, tight cluster-time coincidence, track-quality and vertex-reconstruction requirements, and momentum cuts that retain prompt radiative tridents while suppressing accidentals and converted wide-angle bremsstrahlung. The 2015 and 2016 selections are inherited from the published prompt searches. For 2021, Table 2 distinguishes the upgraded-detector preselection in the full analysis note from the v16 prompt-TC production used for the native 10% v4.9.5 source. The production record establishes $p_{\text{sum}} > 2.8 \text{ GeV}$ and no additional vertex-fit χ^2 requirement. It stores a three-dimensional hit count but does not establish the previously quoted eight-hit two-dimensional electron requirement, so no such requirement is imposed in this description.

Table 2: Comparison of the prompt-resonance selections. The 2015 and 2016 columns follow the published and internal engineering-run summaries. For 2021, the baseline entries follow the full-analysis-note preselection [6, Sec. 4.8, Table 38 and App. A]. The prompt entries list only additional requirements established for the v16 prompt-TC production used for the native 10% v4.9.5 source.

Cut type	2015 (Eng. run 1)	2016 (Eng. run 2)	2021 (native 10%)
Trigger / topology	Pre: pairs-1 trigger; ECAL halves; $y_{e^+}^{\text{clu}} y_{e^-}^{\text{clu}} < 0$. Prompt: same.	Pre: Pair1 trigger bit; opposite ECAL halves; the trigger does not encode track charge. Prompt: same.	Pre: Single-2 or Single-3 positron single-cluster trigger; Single-3 also requires the matched hodoscope condition used by the 2021 trigger menu. Prompt: no additional v16 requirement.
Track momenta	Pre: $p(e^-) < 0.75 E_{\text{beam}}$. Prompt: $p_{\text{sum}} > 0.8 E_{\text{beam}}$ and $p_{\text{sum}} < 1.2 E_{\text{beam}}$.	Pre: $p(e^\pm) < 2.15 \text{ GeV}$; $p_{\text{sum}} < 1.2 E_{\text{beam}}$. Prompt: $p_{\text{sum}} > 1.90 \text{ GeV}$ and $p_{\text{sum}} < 2.4 \text{ GeV}$.	Pre: $p_{\text{trk}} > 0.4 \text{ GeV}$ for each track; FEE veto $p(e^-) < 2.9 \text{ GeV}$; vertex momentum $p_{\text{vtx}} < 4.0 \text{ GeV}$. Prompt: the v16 prompt-TC source requires $p_{\text{sum}} > 2.8 \text{ GeV}$.
Track fit / hits	Pre: loose track-cluster match: $\chi_{\text{trk-clu}}^2 < 10$. Prompt: GBL track $\chi_{\text{trk}}^2 < 40$.	Pre: $\chi_{\text{trk}}^2 < 12$; PID goodness < 10 . Prompt: no additional prompt requirement is documented.	Pre: $\chi_{\text{trk}}^2/\text{ndf} < 20$ and $N_{2\text{D hits}} \geq 10$ for the tracks. Prompt: no additional 2D-hit requirement is established; the available v16 field counts 3D hits.

Cut type	2015 (Eng. run 1)	2016 (Eng. run 2)	2021 (native 10%)
Track-cluster timing	Pre: — Prompt: $ t_{\text{clu}} - t_{\text{trk}} - 43 \text{ ns} < 4.5 \text{ ns}$.	Pre: $ t_{\text{trk}} - t_{\text{clu}} < 6 \text{ ns}$. Prompt: no additional value stated.	Pre: full-note data timing windows: $\Delta(t_{e^-, \text{trk}}, t_{e^+, \text{clu}}) < 6.0 \text{ ns}$, $\Delta(t_{e^+, \text{trk}}, t_{e^+, \text{clu}}) < 5.1 \text{ ns}$, and $\Delta(t_{e^-, \text{trk}}, t_{e^+, \text{trk}}) < 7.8 \text{ ns}$. Prompt: no additional v16 timing requirement.
Cluster-cluster coincidence	Pre: — Prompt: $ \Delta t_{\text{clu-clu}} < 2 \text{ ns}$.	Pre: $ \Delta t_{\text{clu-clu}} < 2 \text{ ns}$. Prompt: $ \Delta t_{\text{clu-clu}} < 1.43 \text{ ns}$.	Pre: no separate offline cluster-cluster coincidence appears in the full preselection table; the Pair-0 hardware trigger has a cluster-pair timing window $ \Delta t < 12 \text{ ns}$ and VTP prescale 100. Prompt: no additional v16 offline coincidence requirement.
Vertex / V_0 quality	Pre: if multiple V_0 remain, choose the best χ^2_{vtx} . Prompt: $\chi^2_{\text{vtx}} < 75$ and $p(V_0) < 1.2 E_{\text{beam}}$.	Pre: target-constrained V_0 candidates. Prompt: no additional prompt χ^2_{vtx} requirement is documented.	Pre: full-note upgraded-detector values: $p_{\text{vtx}} < 4.0 \text{ GeV}$ and $\chi^2_{\text{vtx}} < 20$. Prompt: no additional v16 vertex-fit χ^2 requirement.
Positron cluster energy	Pre: — Prompt: —	Pre: — Prompt: —	Pre: full-note sanity cut $E_{e^+}^{\text{clu}} > 0.2 \text{ GeV}$. Prompt: no additional v16 positron-cluster threshold.
cWAB suppression	Pre: — Prompt: e^+ must have L1 and L2 hits; $d_0(e^+) < 1.1 \text{ mm}$; and $(p_T^- - p_T^+) / (p_T^- + p_T^+) < 0.47$.	Pre: — Prompt: not applied in the final 2016 prompt selection summary.	Pre: no explicit cWAB-suppression cut is listed in the full 2021 preselection table. Prompt: no additional v16 cWAB requirement.
Candidate multiplicity	Pre: unique- V_0 policy: keep the best χ^2_{vtx} . Prompt: enforced by the unique- V_0 logic.	Pre: — Prompt: require a single surviving V_0 candidate.	Pre: exactly one selected good vertex: $N_{\text{vtx}} > 0$ and $N_{\text{vtx}} < 2$. Prompt: no additional v16 multiplicity requirement.

The baseline 2021 entries follow the full analysis note [6, Sec. 4.8, Table 38 and App. A]. The native 10% spectrum comes from the v16 prompt-TC production. The GPR fit adds no event-level cuts. Each campaign enters the analysis through its selected invariant-mass histogram and the detector-response and normalization inputs for the same event category.

The engineering-run notes use *preselection* for the broad reconstruction sample and *prompt selection* for the tighter resonance-search sample. The 2021 source is closer to the latter because the v16 prompt-TC production retains the high- p_{sum} side. Table 2 preserves that distinction without inferring an unrecorded two-dimensional hit requirement.

The 2021 preselection figures below show selected reconstruction variables. The signal simula-

tion has an arbitrary normalization, so only the distribution shapes can be compared, not yields or efficiencies.

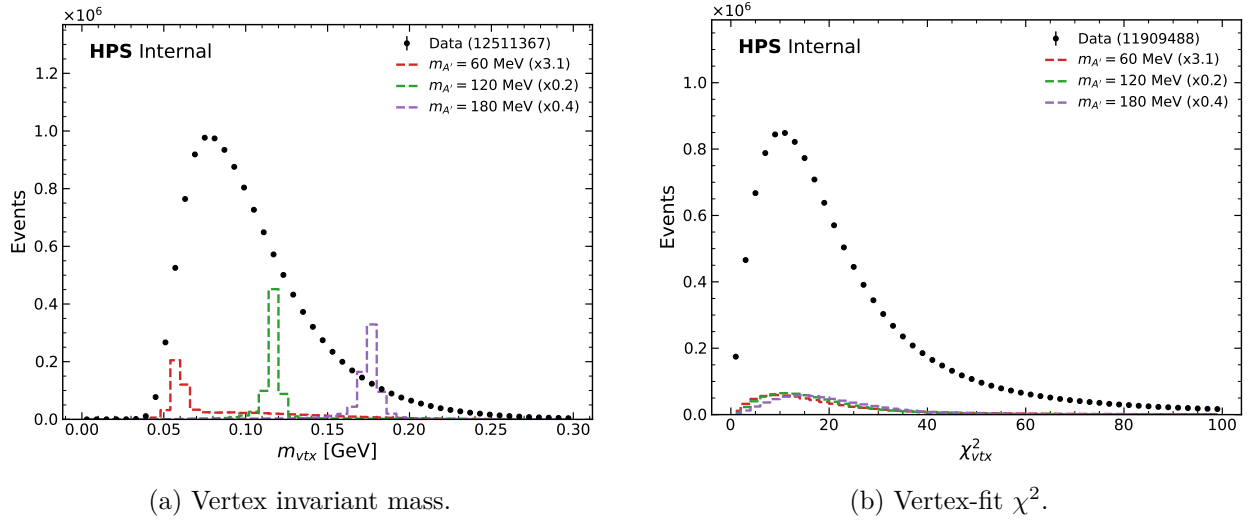


Figure 4: Reconstructed-vertex distributions after the loose preselection, comparing data (points) with the signal MC (filled histogram). The vertex-fit χ^2 is shown as a diagnostic; no vertex-fit χ^2 requirement is applied to the regenerated 2021 GPR input.

3.2 2015 mass resolution

The 2015 signal width is calibrated with Møller data and propagated across mass with simulated A' samples [7]. The Møller peak fixes the detector resolution at a known mass, and the data–MC momentum-resolution difference determines the additional smearing applied to the simulation. Rescaling the fitted A' widths to that data anchor gives

$$\sigma_{2015}(m) = -9.22 \times 10^{-5} + 5.322 \times 10^{-2} m, \quad (1)$$

with m and σ_{2015} expressed in GeV.

The Møller measurement supplies the absolute response, while the simulated mass grid supplies its mass dependence. The associated uncertainty therefore reflects the Møller calibration, the target position, and the residual momentum-response mismodelling [5, 7]. At each test mass this curve sets both the Gaussian signal width and the width of the excluded training interval.

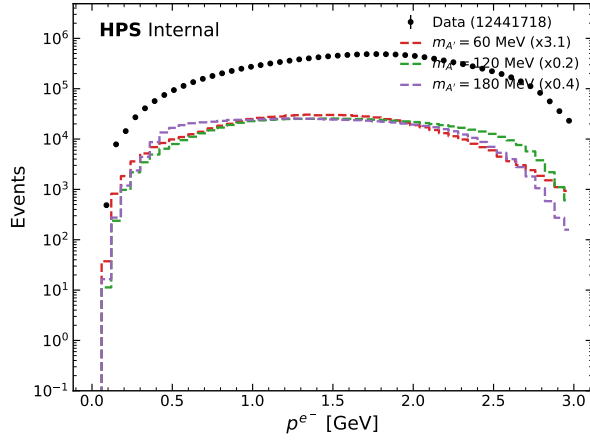
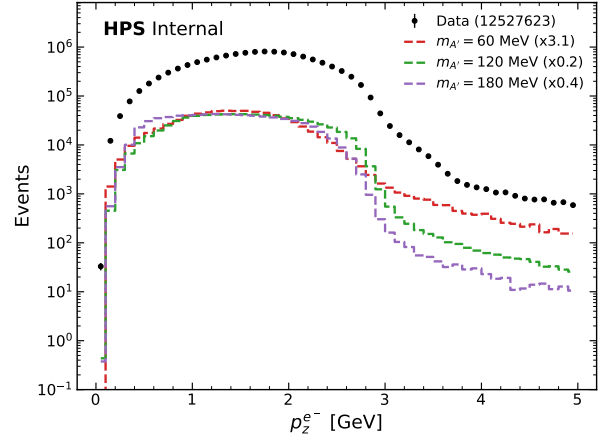
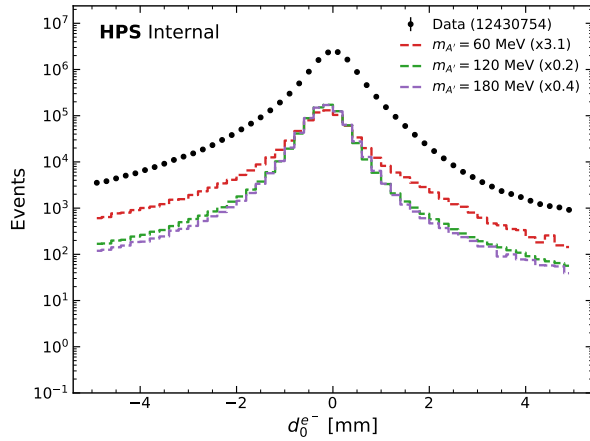
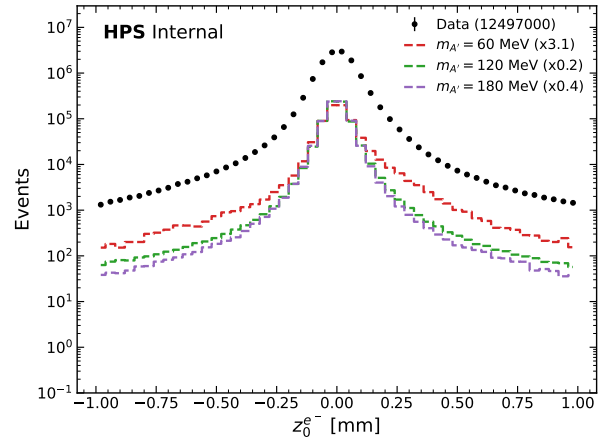
(a) Electron momentum p .(b) Electron longitudinal momentum p_z .(c) Electron transverse impact parameter d_0 .(d) Electron longitudinal impact parameter z_0 .

Figure 5: Electron-track kinematic distributions after the loose preselection, comparing data (points) with the signal MC (filled histogram).

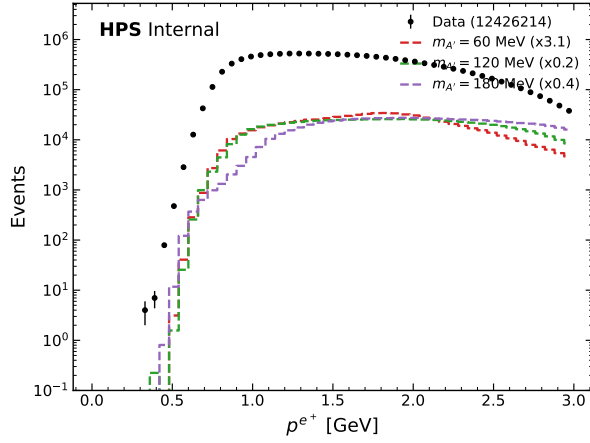
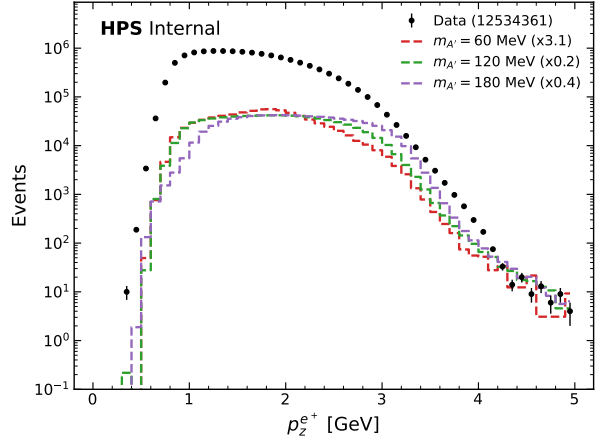
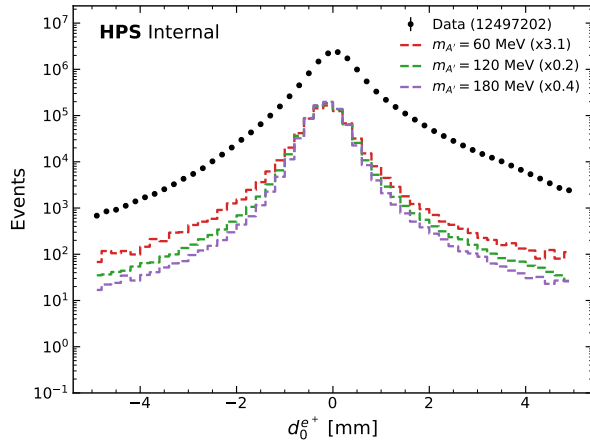
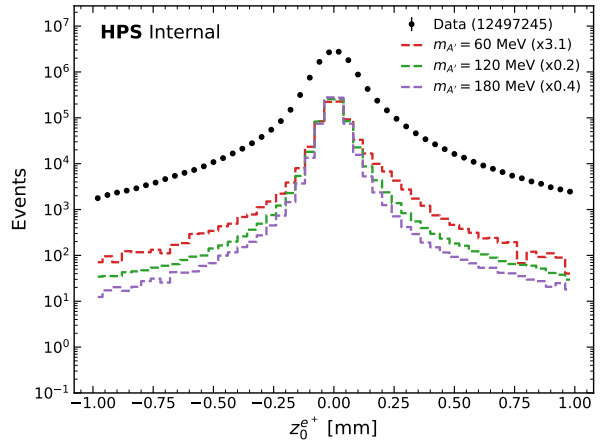
(a) Positron momentum p .(b) Positron longitudinal momentum p_z .(c) Positron transverse impact parameter d_0 .(d) Positron longitudinal impact parameter z_0 .

Figure 6: Positron-track kinematic distributions after the loose preselection, comparing data (points) with the signal MC (filled histogram).

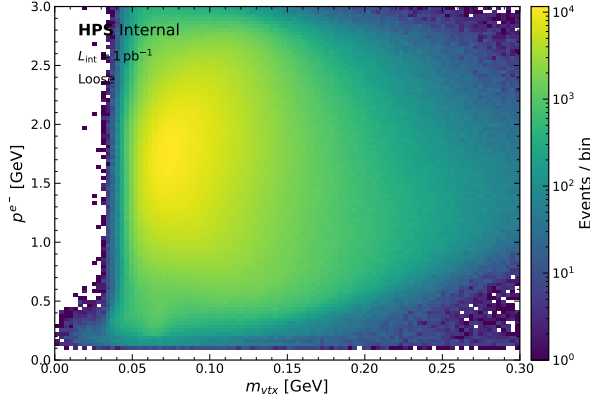
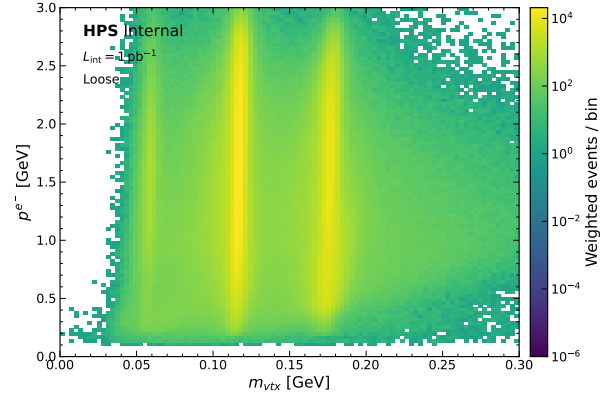
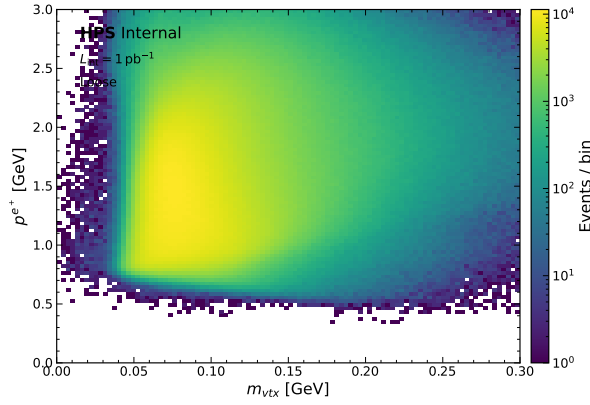
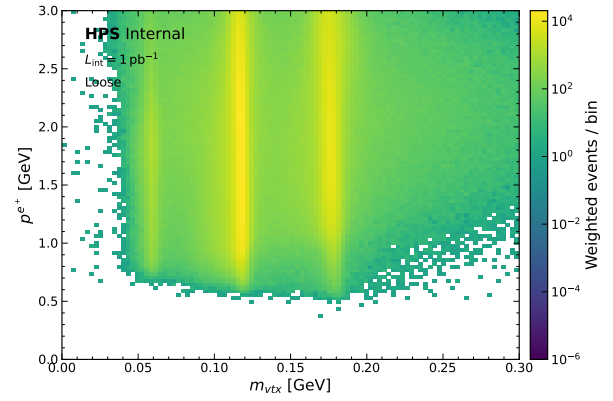
(a) Mass vs. electron p (data).(b) Mass vs. electron p (signal).(c) Mass vs. positron p (data).(d) Mass vs. positron p (signal).

Figure 7: Reconstructed invariant mass versus momentum observables, for data (left) and signal MC (right), after the loose preselection.

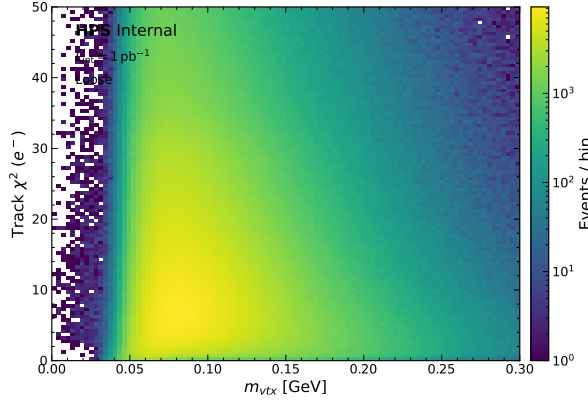
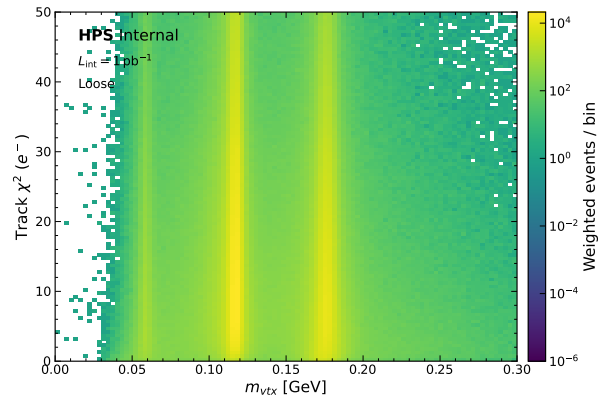
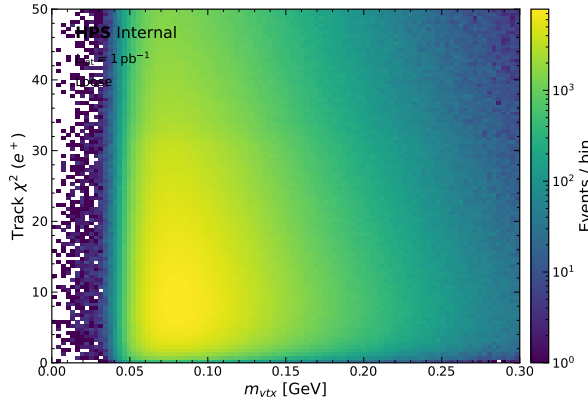
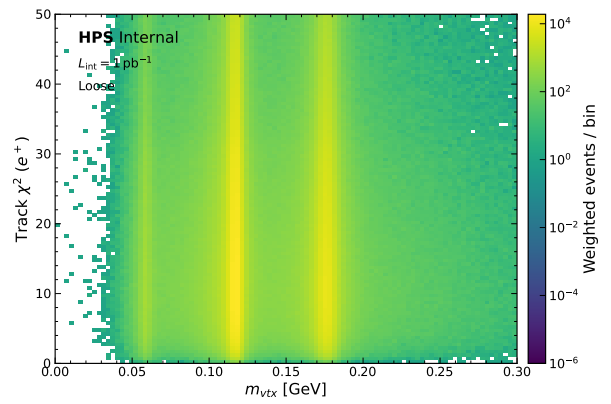
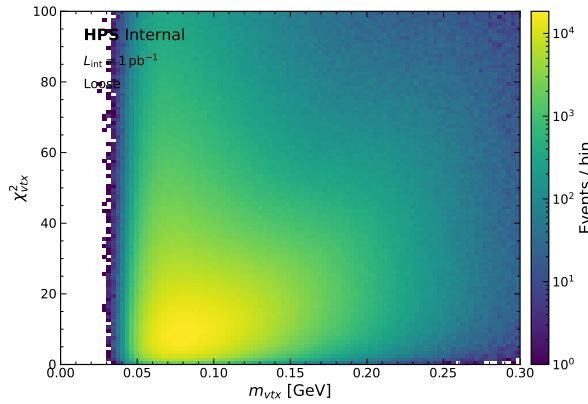
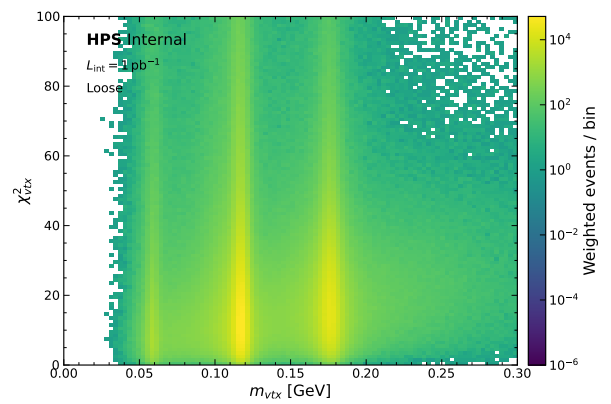
(a) Mass vs. electron χ^2 (data).(b) Mass vs. electron χ^2 (signal).(c) Mass vs. positron χ^2 (data).(d) Mass vs. positron χ^2 (signal).(e) Mass vs. vertex χ^2 (data).(f) Mass vs. vertex χ^2 (signal).

Figure 8: Reconstructed invariant mass versus track- and vertex-fit quality, for data (left) and signal MC (right), after the loose preselection.

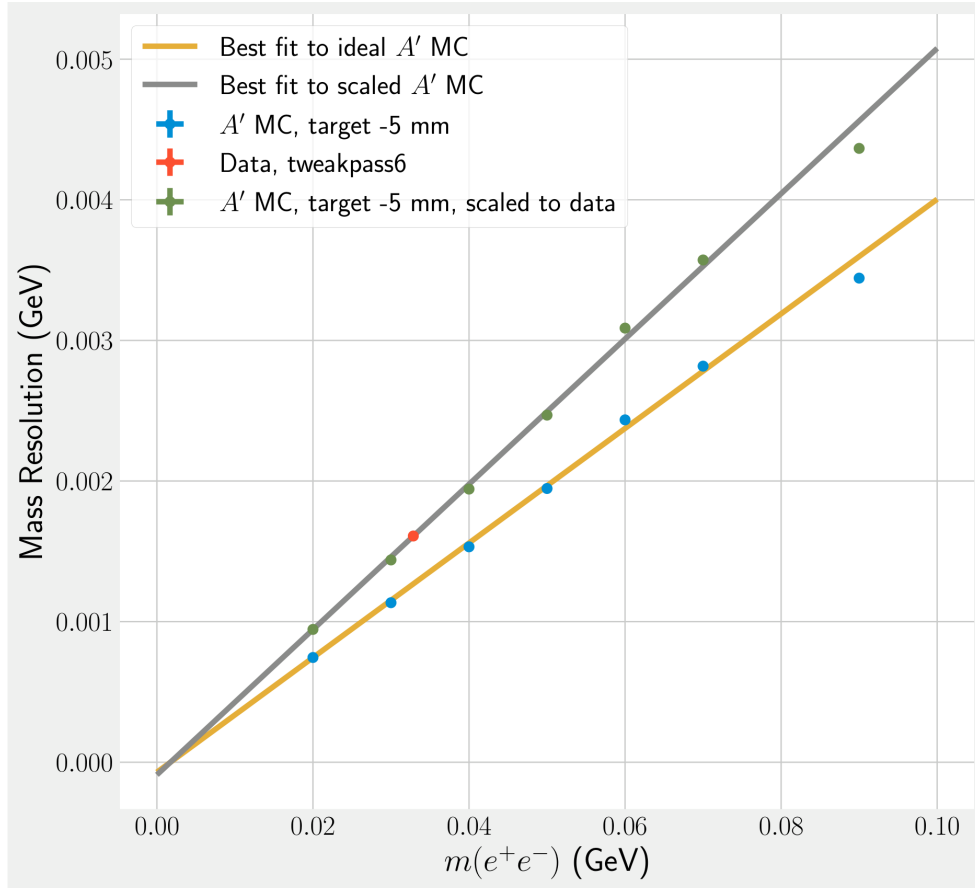


Figure 9: 2015 mass-resolution parameterization from scaled A' MC and Møller data [7]. The Møller point anchors the absolute detector response in data, while the scaled A' simulation supplies the mass dependence across the search window. Because the prompt dark-photon width is negligible, this curve directly sets the resonance width assumed in the search and therefore the local background entering the limit.

3.3 2016 mass resolution

The 2016 calibration follows the same strategy over a broader mass range [3, 2]. Møller events fix the absolute response at 48.5 MeV; full-energy-electron momentum peaks determine the extra smearing needed to match data; and smeared A' samples carry that calibration across the search range. A fourth-order polynomial fit to those samples defines the 2016 signal model,

$$\begin{aligned} \sigma_{2016}(m) = & 3.8 \times 10^{-4} + 4.1 \times 10^{-2}m - 2.7 \times 10^{-1}m^2 \\ & + 3.49m^3 - 11.11m^4, \quad m \leq 0.18 \text{ GeV}, \end{aligned} \quad (2)$$

Earlier extended-range studies continued the polynomial linearly above $m_0 = 0.18 \text{ GeV}$ with slope 0.0239 to avoid its high-mass turnover. The v4 2016 scan ends at 180 MeV, close to the published 39 MeV to 179 MeV interval, so that continuation is not used. The 2015 scan similarly remains near its published 19 MeV to 81 MeV interval, with only the upper endpoint extended to 90 MeV.

The smearing correction reconciles the Møller and simulated A' widths with the observed tracking response [3, 2]. Because the resolution worsens with mass, both the signal template and the excluded training window broaden, increasing the prompt background beneath a narrow signal hypothesis.

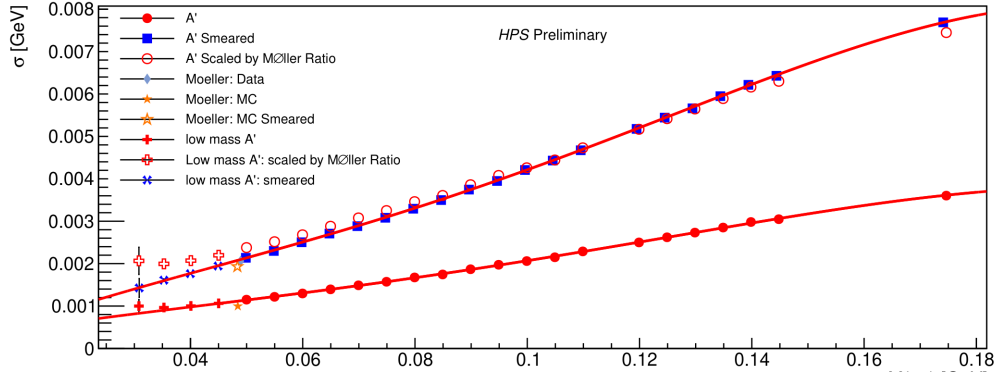


Figure 10: 2016 smeared- A' and Møller-based mass-resolution summary [3]. The smeared A' points show how the data-driven momentum-smearing correction is propagated away from the single Møller calibration point. The worsening resolution with mass broadens the signal window and is one of the main detector effects limiting prompt sensitivity at higher mass.

3.4 2021 mass resolution

The upgraded 2019/2021 tracker requires a separate resolution parameterization. We use the inclusive target-constrained V0 fit without hit requirements shown in Figure 11. Before the final data-MC broadening, the red curve is

$$\sigma_{2021}^{\text{TC,raw}}(m_{\text{MeV}}) = \left(1.4786 - 1.1 \times 10^{-3} m_{\text{MeV}} + 6.87 \times 10^{-5} m_{\text{MeV}}^2\right) \text{ MeV}, \quad (3)$$

where m_{MeV} is the numerical mass value in MeV. The analysis then applies the target-constrained data/MC scale derived from the 2021 smearing study,

$$s_{\text{TC}} = \frac{2.186}{1.745} \simeq 1.25, \quad \sigma_{2021}^{\text{TC,smear}}(m) = s_{\text{TC}} \sigma_{2021}^{\text{TC,raw}}(m). \quad (4)$$

The scale comes from the target-constrained row after hit and curvature (Ω) smearing are applied to the simulation. Equation 4 then sets the 2021 signal width and the corresponding excluded training interval at each mass.

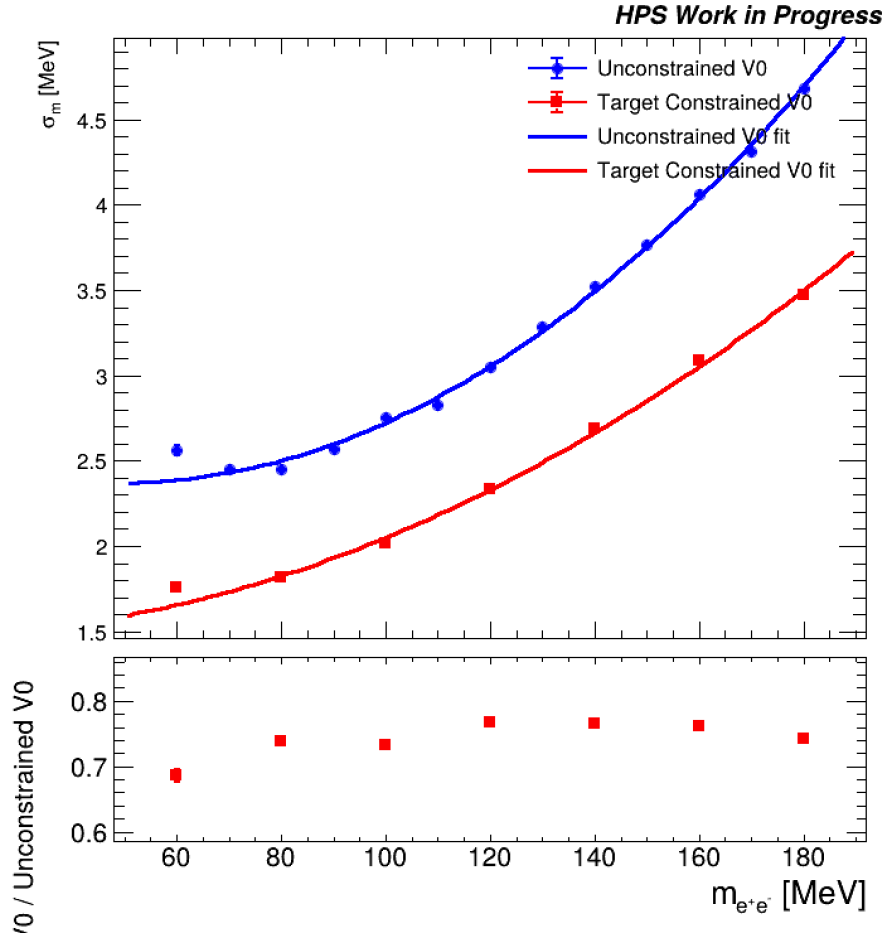


Figure 11: 2021 mass-resolution summary used for the prompt analysis. The red curve is the inclusive target-constrained V0 quadratic fit without hit requirements before the TC data/MC broadening in Eq. 4. The blue curve shows the corresponding unconstrained V0 reference fit included in the underlying study for comparison.

Vertex Type	Hit Smear Only (MeV)	Omega + Hit Smear (MeV)	Data (MeV)
UC	2.329 ± 0.013	2.463 ± 0.014	2.946 ± 0.036
BSC	2.092 ± 0.012	2.193 ± 0.012	2.569 ± 0.035
TC	1.590 ± 0.009	1.745 ± 0.010	2.186 ± 0.031

Figure 12: 2021 Møller run data/MC mass-resolution comparison used to set the smearing scale. Here UC denotes the unconstrained vertex sample, BSC denotes the beamspot-constrained vertex sample, and TC denotes the target-constrained vertex sample. The TC row is the most relevant row for the prompt analysis; comparing the data value 2.186 MeV to the Ω +hit-smear Monte Carlo value 1.745 MeV gives the applied scale factor $2.186/1.745 \simeq 1.25$.

Figure 13 compares the three implemented resolution curves over their scan ranges. The preceding panels show the measurements underlying each parameterization; the combined figure shows the smooth curves used to construct the signal templates and excluded training windows.

Mass-resolution parameterizations over current GPR scan ranges

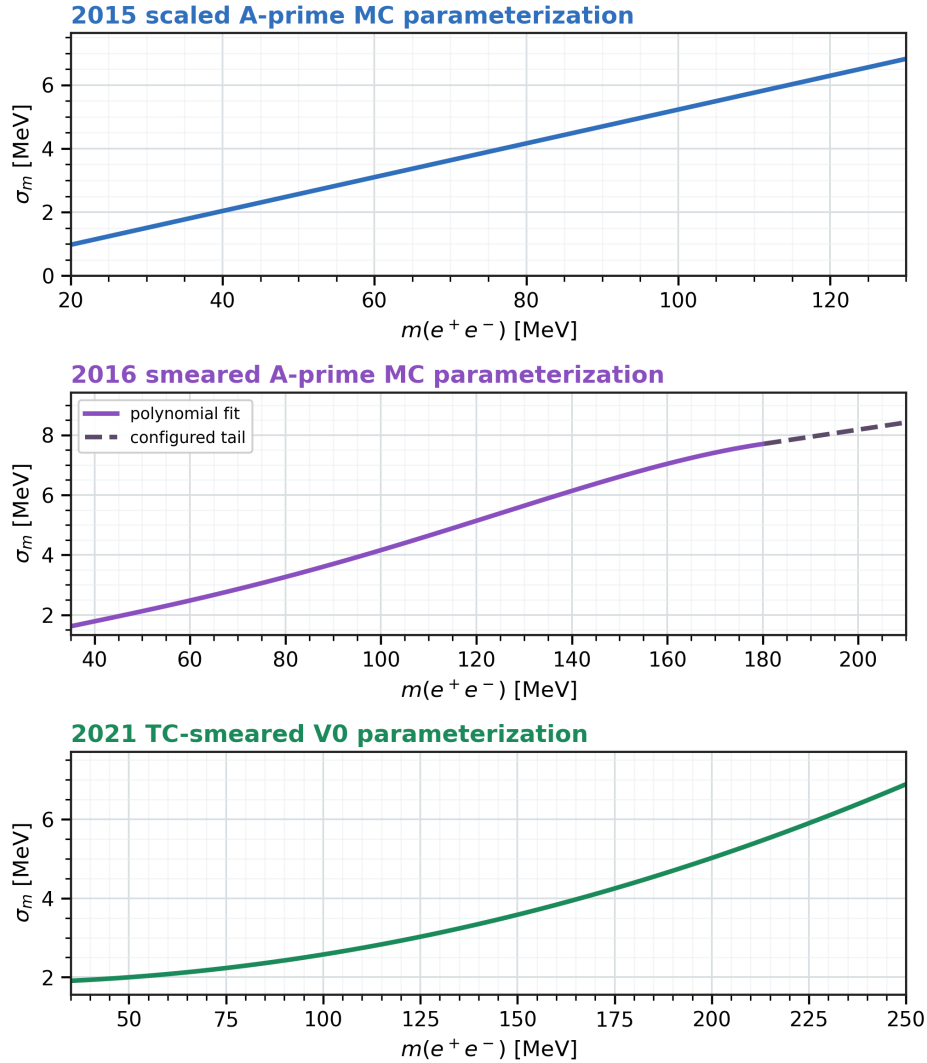


Figure 13: Mass-resolution parameterizations used by the analysis over the adopted scan ranges. The 2016 scan ends at the polynomial boundary of 180 MeV, so its configured high-mass continuation does not enter the v4 result. The 2021 panel shows the target-constrained V0 parameterization after multiplication by the TC smearing scale $s_{\text{TC}} = 2.186/1.745 \simeq 1.25$ and extended to 250 MeV.

3.5 Radiative fraction and the ε^2 normalization

The radiative fraction encodes the fraction of the selected prompt background arising from the purely radiative trident process,

$$f_{\text{rad},d}(m) = \frac{dN_{\gamma^*,d}/dm}{dN_{\text{tri},d}/dm + dN_{\text{wab},d}/dm}, \quad (5)$$

where the numerator is the selected radiative-trident component and the denominator is the full selected prompt background for dataset d [5, 2]. Because heavy-photon production is normalized to the radiative trident rate, f_{rad} is the physics quantity that converts a fitted prompt signal yield

into a limit on the coupling. With this notation,

$$K_d(m) \equiv A_d(\varepsilon^2 = 1) = \frac{3\pi m f_{\text{rad},d}(m) \rho_d(m)}{2\alpha_{\text{EM}}}, \quad A_d = K_d(m) \varepsilon^2, \quad (6)$$

where $\rho_d(m)$ is the selected prompt-spectrum density in dataset d . The engineering-run inputs follow the corresponding prompt searches, while the 2021 normalization remains provisional for the native 10% sample.

3.5.1 2015 radiative fraction

The 2015 prompt search used a constant approximation,

$$f_{\text{rad},2015}(m) = 0.085, \quad (7)$$

which the analysis retains as its 2015 baseline. The dedicated 2015 radiative-fraction study compares the selected radiative-trident component with the total prompt background and adopts the red horizontal line at 8.5% as a conservative description across the search interval. The constant comes from the internal HPS normalization study rather than from the GP fit and is retained for consistency with the published 2015 analysis.

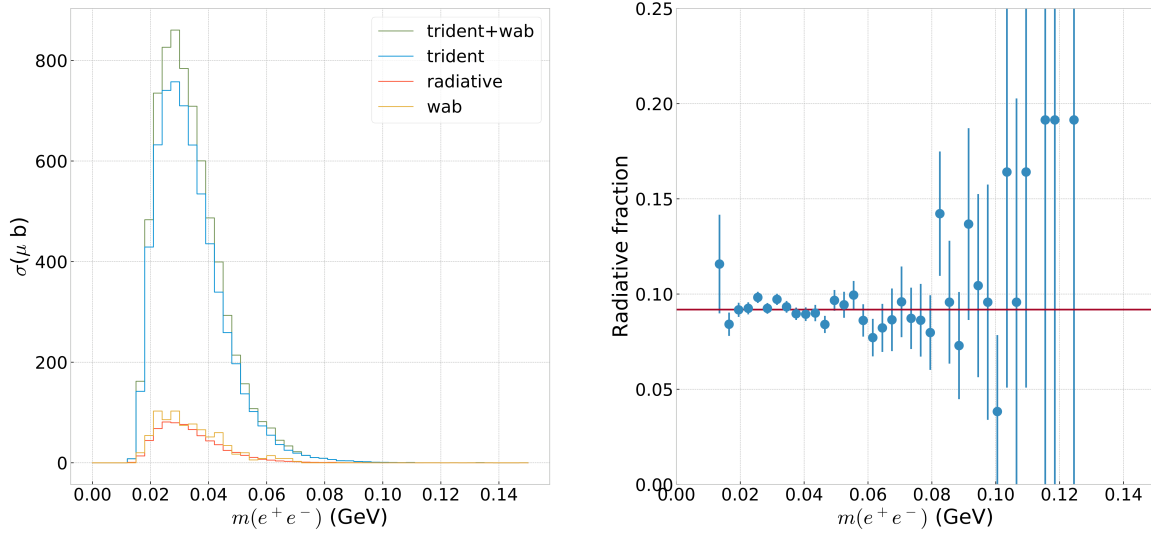


Figure 14: Dedicated 2015 radiative-fraction inputs. The left panel shows the selected cross-section composition, while the right panel shows the radiative fraction used for normalization. The red horizontal line denotes the 8.5% constant adopted as the conservative fit across the 2015 prompt-search dataset. This quantity converts the measured prompt spectrum into the expected A' normalization, so it affects the final ε^2 interpretation rather than the local spectral fit itself.

3.5.2 2016 radiative fraction

The 2016 prompt search used a fifth-order polynomial derived from radiative, trident, and converted-WAB MC after the full prompt selection. Those components determine the mass-dependent conversion from fitted signal yield to ε^2 over the 2016 search range. For the 2016 input, we instead use the constant approximation

$$f_{\text{rad},2016}(m) = 0.05, \quad (8)$$

which is numerically stable over the wider scan range. Figure 15 shows the mass-dependent simulation used in the published determination; the constant used here is a working numerical approximation to that richer model.

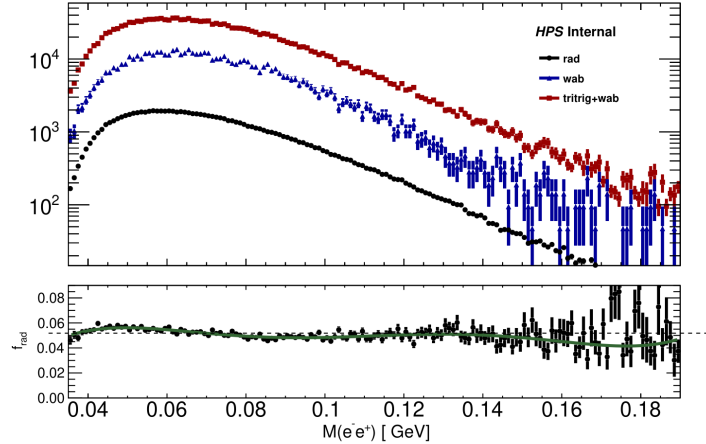


Figure 30: The differential rates in units of MeV^{-1} for all the MC samples which go into the radiative fraction. The rad component is a function of the true mass of the generator level gamma* for the event. The WAB and tritrig components are a function of the invariant mass of the selected V0 candidate. The bottom panel shows the radiative fraction, the green line is the fifth order polynomial fit, and the dotted line is the zeroth order polynomial fit.

Figure 15: 2016 radiative-fraction study from radiative, trident, and converted-WAB MC [3]. The simulation gives an intrinsically mass-dependent normalization, although this analysis uses the constant $f_{\text{rad},2016} = 0.05$ for numerical stability.

3.5.3 2021 radiative fraction

For the native 2021 10% sample, the radiative-fraction input remains provisional. We use

$$f_{\text{rad},2021}(m) = 0.05. \quad (9)$$

The 2021 full analysis note constructs the ratio from selected radiative-trident, tritrig, and WAB samples after preselection and a high-momentum requirement quoted as $p_{\text{vtx}} > 2.8 \text{ GeV}$. A fit over $0.05 < m < 0.25 \text{ GeV}$ gives the constant average $f_{\text{rad}}^{\text{const}} = 0.0496$ [6, Sec. 6.1, Figs. 92–93, Eqs. 25–26]. Thus 0.05 is a reasonable approximation over most of the 2021 search range, but it is not frozen for the final unblinded result. That result will require a direct check of the normalization category against the v16 prompt-TC p_{sum} selection and its threshold convention, with the mass-dependent parameterization retained as a cross-check.

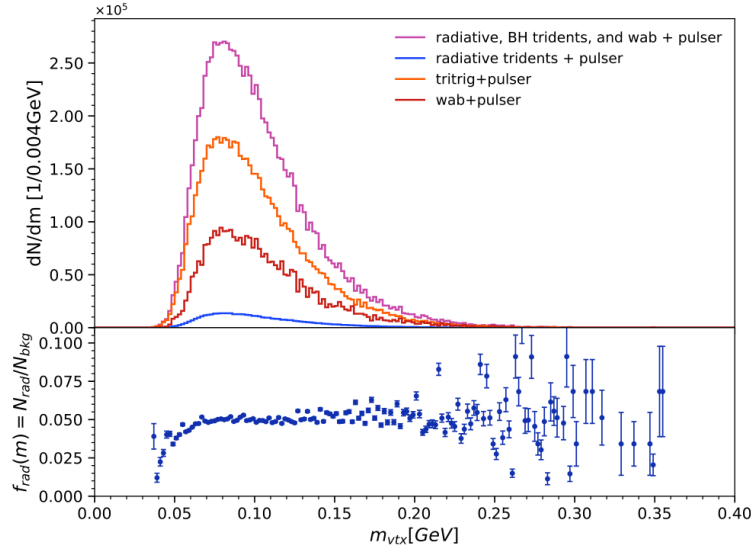


Figure 92: Invariant mass distributions of background samples. The lower panel shows the ratio of radiative tridents to the combined tritrig and WAB samples. This is referred to as the radiative fraction.

Figure 16: 2021 selected-mass distributions used to form the radiative fraction in the full-analysis note [6]. The upper panel shows the selected radiative, tritrig, and WAB components, while the lower panel gives $f_{\text{rad}}(m) = N_{\text{rad}}/N_{\text{bkg}}$.

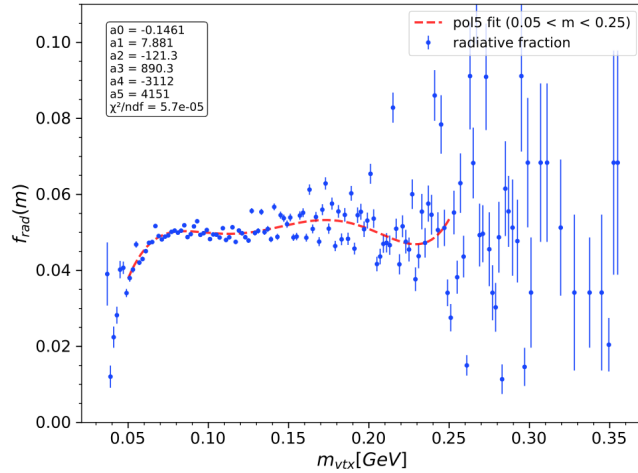


Figure 93: 5th-order polynomial fit to the radiative fraction. The fit range and results are indicated in the figure.

Figure 17: Fifth-order 2021 full-analysis-note fit to the binned radiative fraction [6]. The fitted curve is close to the constant value 0.05 across most of the range used by the 2021 analysis.

3.6 Systematic penalties from the radiative fraction

The analysis applies a data-set-dependent radiative-fraction penalty,

$$f_{\text{rad},d}^{\text{eff}}(m) = (1 - \delta_{f,d}) f_{\text{rad},d}(m). \quad (10)$$

The penalty enters only when signal yield is converted to ε^2 . It changes the normalization of the inferred coupling without altering the blind-window likelihood.

The v4 configuration uses

$$(\delta_{f,2015}, \delta_{f,2016}, \delta_{f,2021}) = (0.070, 0.070, 0.046). \quad (11)$$

For a fixed signal-yield limit, these shifts weaken the coupling-squared limit by factors $1/0.93$ for 2015 and 2016 and $1/0.954$ for 2021. In the combined fit each $K_d(m)$ is reduced by its own amount, so a larger shared ε^2 is required to produce the same yields. The factors are fixed, one-sided normalization shifts; they do not enter the GP interpolation and are not profiled as nuisance parameters.

The 7% engineering-run penalties reflect the several-percent uncertainty in the 2015 radiative and total e^+e^- composition and the corresponding 2016 uncertainty from the selected radiative, trident, and converted-WAB inputs [7, 2]. We retain 7% as a conservative first-order treatment for both engineering runs. The 2021 term uses the upgraded-detector estimate below.

The 2021 full analysis note derives a direct estimate from the generated RAD and WAB cross-section distributions shown in Fig. 18; propagating the quoted RAD and WAB cross-section widths with the relative normalization $R = 0.244$ gives a bounded fractional uncertainty of 3.5% [6, Sec. 9.1, Fig. 161]. The same note assigns an additional 3% preselection/radiative-fraction modeling uncertainty, motivated by WAB-selective mismodeling in the selected background [6, Sec. 9.2]. Treating these two direct radiative-fraction normalization terms as independent gives

$$\delta_{f,2021}^{\text{note}} = \sqrt{(0.035)^2 + (0.030)^2} = 0.046, \quad f_{\text{rad},2021}^{\text{eff}} = 0.954 f_{\text{rad},2021}. \quad (12)$$

The resulting one-sided 2021 penalty is 4.6%, corresponding to a multiplicative factor of 0.954. The 2015 and 2016 terms remain at 7%.

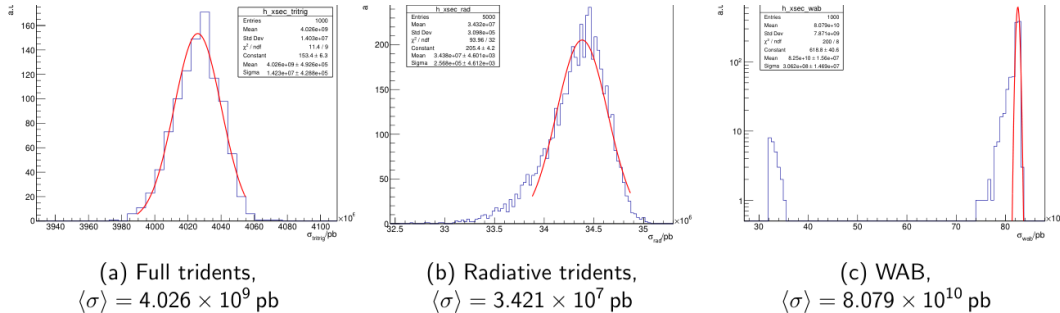


Figure 16: Distribution of generated cross sections for background MC samples.

Figure 18: Generated cross-section distributions for the 2021 tritrig, radiative, and WAB background MC samples [6]. These distributions set the cross-section normalization inputs for the 2021 radiative-fraction calculation.

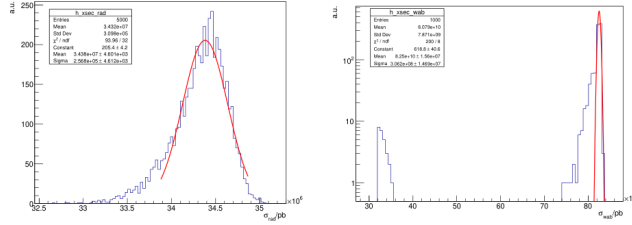


Figure 161: **Left:** Distribution of cross sections for radiative events, **Right:** Distribution of cross sections for WAB events.

Figure 19: RAD and WAB cross-section distributions used in the 2021 note’s radiative-fraction cross-section uncertainty propagation [6]. Together with the relative normalization $R = 0.244$, these inputs lead to the 3.5% MC cross-section contribution.

The 4.6% term covers only the radiative-fraction normalization. The broader 2021 displaced-search uncertainty model in Fig. 20 also includes mass resolution, target position, and tight-selection effects that must be revisited before applying it to the full data set [6, Sec. 9, Table 58]. Those signal-yield and acceptance uncertainties are not included in Eq. (10).

Source	Percentage
MC Cross Section	3.5%
Pre-selection Uncertainty	3%
Invariant Mass Resolution	6.5%
Target Position Uncertainty	9%
MC-Data Tight Selection Uncertainty (Intuition)	2%
MC-Data Tight Selection Uncertainty (ANN)	4%

Table 58: Table of (expected) uncertainties for the displaced vertex search

Figure 20: Broader 2021 full-analysis-note systematic summary [6]. Only the MC cross-section and preselection entries are folded into the radiative-fraction-only penalty in Eq. (12); the other entries remain separate signal-yield or acceptance effects.

4 Methodology

4.1 From piecewise windows to fixed-support GP fits

The original HPS GPR studies explored piecewise mass windows, redefined the sidebands at each hypothesis, and considered several background profiles. They established that a GP could interpolate the smooth trident spectrum, but the window-by-window fits also produced less smooth limit curves, increased sensitivity to local fit boundaries, and made a coherent multi-dataset combination difficult. The present workflow instead fits the GP separately at each hypothesis on a fixed, dataset-specific support, excluding only a narrow mass-dependent training region around the test mass. The support extends beyond the signal-search interval, but need not include empty turn-on bins or the full ROOT-histogram axis. This construction gives a smoother background prediction that can be validated and used in a combined likelihood with a shared parameter of interest.

4.2 Search interval and GP fit support

The search interval answers where a resonance hypothesis is tested. The GP fit support answers which surrounding bins constrain the smooth background at that hypothesis. Using the same endpoints for both definitions leaves the first and last test masses with a one-sided training sample, even though the input histogram contains usable data beyond the search. The accepted v4.2 three-campaign analysis uses the established 19–90, 39–180, and 50–250 MeV search intervals, with 14–135, 30–210, and 40–300 MeV fit supports for 2015, 2016, and 2021, respectively. The separate v4.9.5 2021-only pass keeps the 50–250 MeV search interval and changes that dataset’s fit support to 36–300 MeV.

These endpoints were declared before the v4 observed scan. The 2015 domain and the upper 2016 edge follow the earlier functional-form validation fits. The 2016 lower edge is placed at 30 MeV rather than the historical 35 MeV fit edge because the exact rebin-five geometry matters: at a 39 MeV hypothesis the blind interval begins at 35.063 MeV, while the first bin center from a 35 MeV support is 35.125 MeV. A nominal 35–210 MeV support would therefore still contain no low-side training-bin center at the lower search endpoint. The adopted domains retain at least 12 and 20 low-side training bins at the 2015 and 2016 lower endpoints, and at least 138 and 51 high-side training bins at their upper endpoints. For the accepted v4.2 geometry, the corresponding 2021 counts are 9 and 55. The bin-count audit is applied to every active dataset and mass hypothesis, rather than inferred only from continuous support boundaries.

For 2021, v4.9 distinguished the support used to generate counts from the support retained in a GP fit. Counts were generated once over 30–300 MeV; paired extractions then used 30–300 and 40–300 MeV supports. This same-toy construction isolated the effect of the lower edge, but did not freeze a new analysis support.

Version 4.9.5 tested lower edges at 32, 34, 36, 38, and 40 MeV, with 30 MeV retained as an ineligible control, on the same 100 archived native-10% conditional backgrounds. The tests used masses of 55, 60, 65, and 70 MeV and injected strengths of $z = 0, 2, 5$. Under the original strict protocol, a candidate failed if any injected cell had $|\bar{p}| \geq 0.5$ and its two-sided 90% Student- t interval excluded zero; no edge passed. The practical rule adopted after that phase required at least 9 of the 12 mass-strength means, including at least 3 of the 4 background-only means, to have $|\bar{p}| < 0.75$, with none at or above 1.25; the technical gates were unchanged. Both 36 and 38 MeV qualified. The declared minimax tie rule selected 36 MeV because its score lay within the 0.10 tie margin and it retained more support. The continuation and the full 100-background summary passed without further retuning. This conditional result froze the 36–300 MeV support for the

v4.9.5 2021-only observed pass; it is not a direct coverage result and does not revise the accepted v4.2 combined-analysis card. The four configuration states are summarized in Table 3.

Analysis state	2021 mass domains	Fixed settings	Interpretation
v4.2 analysis card	search 50–250; GP support: 40–300 MeV	Rebin five; log-mass/log-count preprocessing; $k_{\min} = 1.1$, $k_{\max} = 15$; $2.25\sigma_m$ extraction and training exclusion; signed amplitude and asymptotic 90% CL_s	Historical inference baseline. Its accepted observed result is not changed or recomputed by the support study.
v4.5 matched-refit contract	inherits the v4.2 domains	Refit the GP for the matched background-only reference and every injected state; define $\sigma_{A,\text{ref}}$ from that reference; use $z = 0, 1, 3, 5$ and retain signed pulls	Defines the toy construction only; it neither changes the production settings nor overrides the fit support.
v4.9 support diagnostic	search 50–250; generated spectra: 30–300 MeV; paired GP supports: 30–300 and 40–300 MeV	The paired extractions use identical cached counts, 25 backgrounds per generating source, and the same v4.2/v4.5 fit rules	Completed conditional diagnostic. The lower-edge effect is isolated by a same-toy, full-support control; no new production support is frozen.
v4.9.5 2021-only card	search 50–250; GP support: 36–300 MeV	1 MeV grid; $k_{\min} = 1.1$, $k_{\max} = 15$; $2.25\sigma_m$ extraction and training exclusion; $1.64\sigma_m$ density window; signed amplitude; asymptotic 90% CL_s with $\alpha = 0.10$	Frozen for the native-10% 2021 observed pass after the declared continuation. The support decision is conditional, is not direct coverage, and does not update the v4.2 three-campaign card.

Table 3: Analysis states from v4.2 through v4.9.5. Search interval, toy-generating support, and GP fit support are separate quantities. The accepted v4.2 combined result and the v4.9.5 2021-only observed pass therefore retain distinct 2021 supports.

4.3 Observable, scan grid, and blinding

The input observable is the reconstructed e^+e^- invariant-mass histogram for each dataset. Throughout this section, d labels the dataset, m denotes the reconstructed e^+e^- invariant mass, m_0 is the test-mass hypothesis, and $\sigma_d(m)$ is the dataset-specific mass resolution introduced in Section 3. The default scan step in the package is

$$\Delta m = 1 \text{ MeV}, \quad (13)$$

and the blind window for a mass hypothesis m_0 is centered on the detector resolution,

$$\mathcal{B}_d(m_0) = [m_0 - n_{\text{blind}}\sigma_d(m_0), m_0 + n_{\text{blind}}\sigma_d(m_0)]. \quad (14)$$

The package exposes the extraction half-width and GP training exclusion as independent configuration values. Earlier injection and GP-toy validation studies used $n_{\text{blind}} = 1.64$ with the same $1.64\sigma_d$ training exclusion. The current observed-limit and refmatched closure baseline documented

later in this note uses a wider $2.25\sigma_d$ blind/extraction window with the GP training exclusion also set to $2.25\sigma_d$. For an ideal Gaussian line shape, the $1.64\sigma_d$ interval contains

$$f_{\text{blind}}^{\text{ideal}}(1.64) = \text{erf}\left(\frac{1.64}{\sqrt{2}}\right) \simeq 0.899, \quad (15)$$

while the $2.25\sigma_d$ interval contains

$$f_{\text{blind}}^{\text{ideal}}(2.25) = \text{erf}\left(\frac{2.25}{\sqrt{2}}\right) \simeq 0.976. \quad (16)$$

This point matters conceptually and numerically: the current code does *not* renormalize the signal template inside either blind window. Instead, it keeps the full Gaussian normalization and lets only the blinded subset enter the local likelihood.

The notation $\mathcal{B}_d(m_0)$ denotes the set of histogram bins that lie inside the blinded interval for dataset d at test mass m_0 . In the current production configuration there is no additional gap between the extraction window and the GP training sample: when the extraction and training-exclusion half-widths are equal, the training exclusion starts immediately outside $\mathcal{B}_d(m_0)$. A wider exclusion, for example $1.96\sigma_d$ instead of $1.64\sigma_d$, reduces signal leakage into the training region but also increases the interpolation distance and therefore the GP predictive uncertainty. In the simple background-dominated proxy $S/\sqrt{B} \propto F(k)/\sqrt{k}$ with $F(k) = \text{erf}(k/\sqrt{2})$, the optimum lies near $k \simeq 1.4$, so the historical $1.64\sigma_d$ extraction was close to the pure-sensitivity optimum. Section 5 then motivates the current $2.25\sigma_d$ blind/extraction choice as a signal-recovery improvement after full-refit pseudoexperiments showed appreciable training-region absorption at narrower exclusions.

The bin assignment is implemented with the fixed histogram bin centers, not by splitting partly overlapping bins. Let $c_i = (e_i + e_{i+1})/2$ be the center of bin i . The full bin is included in the blinded likelihood when

$$m_0 - n_{\text{blind}}\sigma_d(m_0) \leq c_i \leq m_0 + n_{\text{blind}}\sigma_d(m_0), \quad (17)$$

and the full bin is included in the GP sideband training sample only when its center lies strictly outside the training-exclusion interval. Thus a bin whose edge overlaps the blind boundary is treated by its center: if the center is inside the blind interval, the entire bin is blinded and enters the local likelihood; if the center is outside, the bin remains available for training unless it is removed by a wider training-exclusion interval. No fractional observed count is assigned to both regions. This same center mask is used consistently for the observed counts, the GP prediction points, the covariance slice, and the blinded-bin signal-template slice.

The finite binning is therefore part of the definition of the numerical result. The Gaussian signal template is integrated over the full bin edges in Eq. (33), but the selected set of bins changes discretely when a boundary passes a bin center. Consequently the quoted local significance and ε^2 limit are not meaningful to a precision finer than the mass binning and scan grid used to define these masks. The production toys and validation plots use the same binning convention, so this edge-bin effect is accounted for by applying the identical discrete procedure to data, pseudoexperiments, background predictions, signal templates, and yield-to-coupling conversions.

4.4 Log-space preprocessing and the choice $\alpha = 1/y$

Following recent HEP GPR background-modeling studies [8, 9], the package fits the GP in transformed coordinates:

$$x = \log m, \quad z = \begin{cases} \log y, & y > 0, \\ 0, & y = 0, \end{cases} \quad (18)$$

where y is the observed bin count. The log transformation is used for two related reasons. First, the prompt trident spectrum spans a wide dynamic range, and modeling z rather than y turns multiplicative shape variations into approximately additive ones. Second, $x = \log m$ makes equal distances in the GP coordinate correspond to equal fractional changes in mass, which is the relevant scale once the detector resolution is written in the transformed coordinate as $\sigma_x(m) \simeq \sigma_d(m)/m$. Barr and Liu emphasize that applying logarithms to both axes improves convergence and performance for wide, steeply falling spectra [8]. The GP regression step uses scikit-learn’s GaussianProcessRegressor estimator [10]; the surrounding HPS analysis code supplies the histogram handling, mass-dependent sideband masks, signal templates, profile likelihood, and CL_s limit construction.

The GP is therefore used to model an unobserved smooth mean function $z_{\text{true},d}(x)$ for the log-count spectrum. In that language, “latent” means only that the smooth log-rate is not observed directly; it is inferred from the histogram counts through the transformed coordinates and the heteroscedastic variance term. The GP mean vector is the vector of latent log-count means evaluated at the selected training-bin coordinate. For Poisson counts $Y_i \sim \text{Pois}(\lambda_i)$, the delta method gives $\text{Var}[\log Y_i] \approx \text{Var}[Y_i]/E[Y_i]^2 \approx 1/\lambda_i$, which motivates the plug-in choice $\alpha_i \approx 1/y_i$ for occupied bins. A heteroscedastic diagonal term α_i is then added to the training covariance,

$$\alpha_i = \begin{cases} 1/y_i, & y_i > 0, \\ 1, & y_i = 0, \end{cases} \quad (19)$$

$$K_{\text{obs}} = K(X, X) + \text{diag}(\alpha), \quad (20)$$

so that each α_i plays the role of an observation-variance term for bin i . This is the quantity that GaussianProcessRegressor adds to the diagonal during fitting, so it is treated here as a variance model rather than an arbitrary tuning parameter.

This is a deliberate departure from the earlier Barr–Liu-inspired draft parameterization. In the present analysis, α_i is treated as the latent-space variance assignment associated with the transformed counts, while avoiding the extra $\log y$ weighting explored in the first rough drafts. For empty bins the adopted floor $\alpha_i = 1$ is a unit variance in log-count space, not a pseudo-count of one event. It simply prevents zero-count bins from receiving ill-defined or disproportionate weight after the log transformation. In this high-statistics search such bins are rare, so the floor acts mainly as a conservative legacy-stable safeguard rather than as a dominant source of uncertainty.

The distinction from Ref. [8] is more specific than a difference in kernel choice. Barr and Liu also use a Constant $\times$ RBF kernel and optimize its amplitude and length scale. Their independent methodological study uses a spectrum derived from a published CMS search; it is not a CMS Collaboration GPR result. In addition to optimizing the kernel, they treat the diagonal array as tunable regularization. For occupied bins their nominal prescription is

$$\alpha_i^{\text{BL}} = \frac{\sqrt{y_i}}{y_i \log y_i} = \frac{1}{\sqrt{y_i} \log y_i}, \quad (21)$$

and fixed regional multipliers f_i are then applied to α_i^{BL} . Reducing the first eleven multipliers to 0.1 raised the fraction of 100 background-only pseudoexperiments passing their BumpHunter criterion from 51% to 87%, but localized residual structures remained; their high-luminosity benchmark subsequently extended the reduced-multiplier region to the first twenty bins [8]. The study therefore demonstrates that diagonal tuning can improve a particular benchmark, while also showing that the tuned region is sample- and regime-dependent.

The active HPS configuration applies no regional multiplier to the early bins and fixes Eq. (19) as the delta-method variance model. This choice matters because scikit-learn’s alpha parameter has

both an observation-noise interpretation and a numerical Tikhonov-regularization interpretation [11]. HPS adopts the former and places its separately validated smoothness control in the allowed length-scale domain. These are complementary controls: changing α_i changes the leverage of bin i on the covariance diagonal, whereas changing ℓ changes the off-diagonal correlation range of the latent function. No matched study presented here establishes that one strategy is intrinsically superior to the other.

4.5 Kernel choice and distinct regularization roles

The baseline kernel in the current package is a single ConstantKernel $\times$ RBF model,

$$k(x, x') = C \exp \left[-\frac{(x - x')^2}{2\ell^2} \right], \quad (22)$$

with broad amplitude bounds and a resolution-scaled RBF length-scale domain. This is the same simple kernel class used in the methodological studies of Refs. [9, 8], but the surrounding noise and validation choices need not be the same. The two quantities optimized in every HPS sideband fit are C and ℓ . The ConstantKernel value $C = k(x, x)$ is the pointwise prior variance of the latent log-rate, so $\sqrt{C}$ is its prior standard deviation; C is not a fitted event-count normalization. The length scale ℓ controls how quickly covariance falls with separation in $x = \log m$, and therefore how rapidly the background function may bend. The fixed α_i values are not a third kernel parameter and there is no optimized WhiteKernel in the baseline. They enter only through the observation-covariance diagonal.

Figure 21 separates these roles graphically. This separation is useful when discussing regularization: changing C rescales the covariance amplitude, changing ℓ changes the correlation geometry and smoothness of the latent function, and changing α_i changes how strongly one observation constrains that function.

$C \times \text{RBF}$ plus observation noise: three different roles

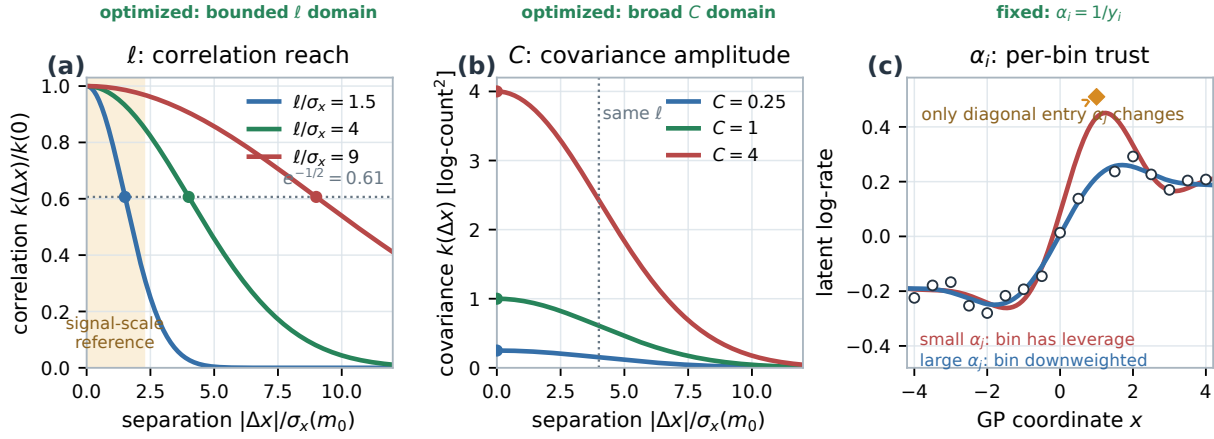


Figure 21: Division of labor in the baseline GPR covariance. Panel (a) shows the normalized RBF correlation for three illustrative values of $\ell/\sigma_x(m_0)$; at a separation equal to ℓ , every curve has correlation $e^{-1/2} \simeq 0.61$. The length scale therefore changes the horizontal range over which bins communicate. Panel (b) varies C at fixed ℓ . It changes the covariance amplitude, with pointwise prior standard deviation $\sqrt{C}$, without changing the normalized correlation range. Panel (c) holds the kernel fixed and changes one diagonal variance α_j : a smaller value gives the highlighted bin more leverage, while a larger value downweights it. HPS optimizes C and ℓ on the sidebands within configured bounds, but fixes α_i from the count-based variance rule. All curves and observations are schematic and are not an HPS fit.

Earlier studies explored more complicated kernels and piecewise procedures, but those variants are no longer the production baseline. The Constant $\times$ RBF model is the analysis kernel; exploratory Constant $\times$ Matérn($\nu = 3/2$) and piecewise-kernel studies remain robustness checks. None provides a production-matched alternative-kernel coverage result for the v4.2 analysis.

4.6 Resolution-scaled length-scale parameterization and bounds

Because the GP is trained in $x = \log m$, detector resolution must be translated into log-mass units. The package defines

$$\sigma_x(m) = \log(m + \sigma_d(m)) - \log m \simeq \frac{\sigma_d(m)}{m} \quad (\sigma_d \ll m), \quad (23)$$

and uses this local physical scale to set the RBF bounds. Under the local, resolution-scaled policy,

$$\ell_{\min,d}(m) = k_{\min,d}\sigma_x(m), \quad \ell_{\max,d}(m) = k_{\max,d}\sigma_x(m). \quad (24)$$

For the RBF covariance used here,

$$k_{\text{RBF}}(x, x') = \sigma_f^2 \exp\left[-\frac{(x - x')^2}{2\ell^2}\right], \quad \frac{k_{\text{RBF}}(x, x + \ell)}{k_{\text{RBF}}(x, x)} = e^{-1/2} \simeq 0.61. \quad (25)$$

Thus ℓ is a covariance-decay scale in the GP input coordinate: larger values favor smoother background functions, while smaller values admit more rapid variation [12, 13]. The ratio $\ell/\sigma_x(m)$

expresses that nuisance-model smoothness relative to the width of the narrow signal hypothesis. It is not, without an independently validated mapping, a detector resolution measurement or a physical background correlation length. The fitted ℓ_{opt} is the log-marginal-likelihood optimum conditional on the kernel, data support, sideband mask, noise model, and configured optimizer domain. Consequently an optimum at ℓ_{max} establishes only that the numerical constraint is active; it neither measures a physical scale at the boundary nor locates the unconstrained optimum. The implementation optimizes log-transformed kernel hyperparameters within these configured bounds [11].

For a fixed dataset d and mass hypothesis m_0 , let $\mathbf{z}_{\text{sb}}$ contain only the transformed sideband observations and define

$$K_y(C, \ell) = C R_\ell(X_{\text{sb}}, X_{\text{sb}}) + \text{diag}(\boldsymbol{\alpha}_{\text{sb}}). \quad (26)$$

The optimized pair maximizes

$$\log p(\mathbf{z}_{\text{sb}} \mid X_{\text{sb}}, C, \ell) = -\frac{1}{2} \mathbf{z}_{\text{sb}}^T K_y^{-1} \mathbf{z}_{\text{sb}} - \frac{1}{2} \log |K_y| - \frac{n_{\text{sb}}}{2} \log(2\pi). \quad (27)$$

The quadratic term rewards agreement with the sideband observations, while the determinant term accounts for the covariance volume used to obtain that agreement. The implementation searches in the logarithms of C and ℓ with multiple optimizer restarts. This is an empirical-Bayes, or type-II maximum-likelihood, selection of the covariance parameters; it is not a profile of ℓ as a frequentist nuisance parameter in the subsequent local CL_s likelihood.

Figure 22 shows why the optimum is specific to a dataset and mass hypothesis. Moving m_0 changes which bins are excluded from training and changes the resolution-scaled domain through $\sigma_x(m_0)$. The same dataset can therefore return a different $(C_{\text{opt}}, \ell_{\text{opt}})$ at neighboring hypotheses even though the kernel family is unchanged. An interior maximum means only that the configured range is nonbinding for that fit. A maximum on a boundary means that the range is active, while a long shallow likelihood ridge means that the parameters are weakly identified. None of these cases turns ℓ_{opt} into a direct measurement of detector resolution or a physical background correlation length.

What an optimized length scale means at a mass hypothesis

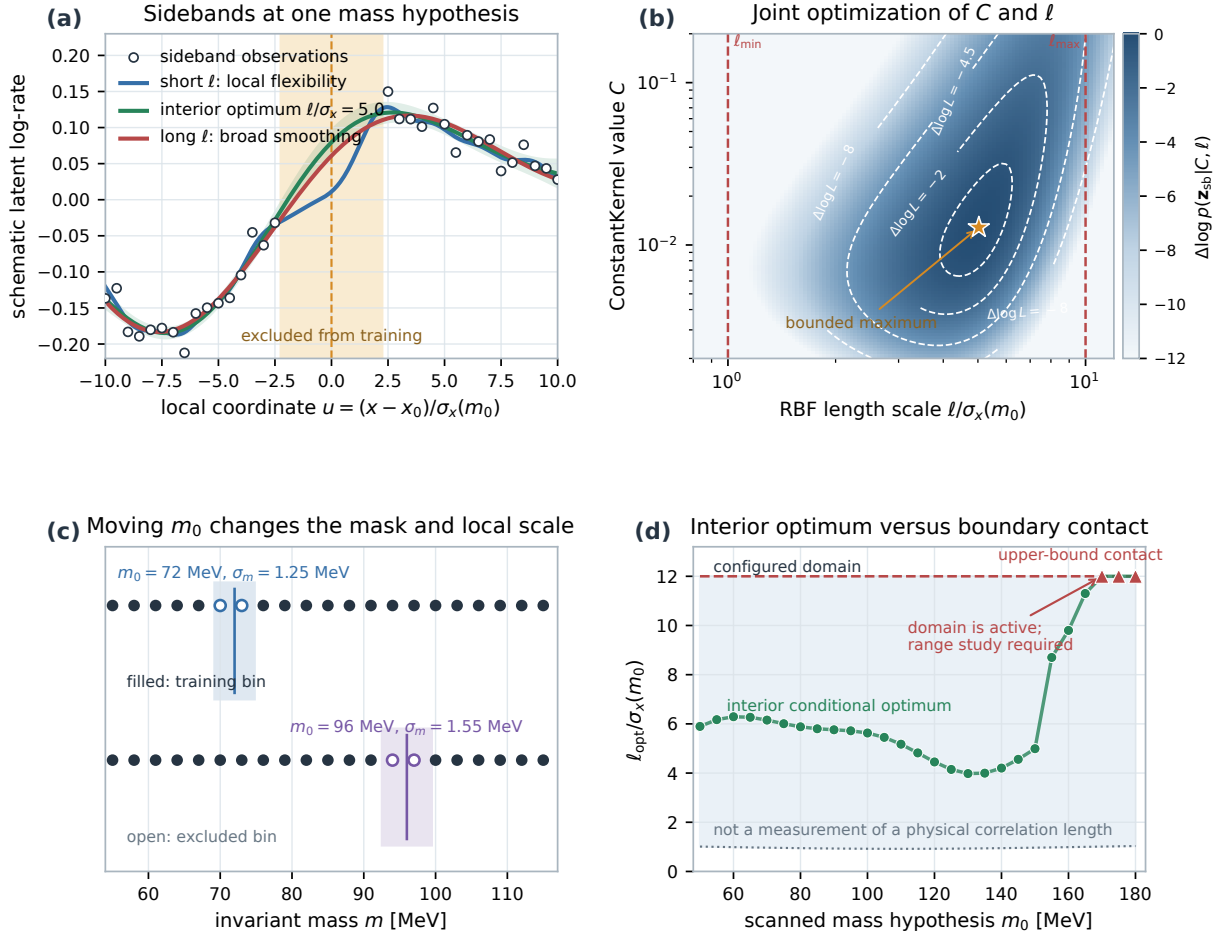


Figure 22: Meaning of per-mass-hypothesis hyperparameter optimization. Panel (a) shows one schematic sideband problem in $u = (x - x_0)/\sigma_x(m_0)$. The shaded $2.25 \sigma_x$ interval is excluded from training; the three curves illustrate the different interpolation behavior allowed by short, interior, and long candidate length scales. Panel (b) shows the joint log-marginal-likelihood surface for the same synthetic sidebands. The star is the bounded maximum in (C, ℓ) , and the elongated contours expose their possible tradeoff. Panel (c) shows that moving m_0 changes both the sideband mask and the local resolution scale that defines the allowed ℓ interval. Panel (d) illustrates how a scan records an interior optimum differently from repeated upper-bound contact: the former is a conditional best fit within a nonbinding range, whereas the latter diagnoses an active numerical domain and motivates a predeclared range and closure study. The figure is explanatory only; it contains no HPS observations or fit results.

This distinction determines how the hyperparameter study enters a physics result. If the allowed range forces ℓ too low, the GP retains unnecessary signal-scale flexibility and can absorb part of a localized excess; if the fitted background is forced too smooth, broad mismodeling can instead leave a biased local residual. Raising only the upper bound permits, but does not require, a smoother solution. A larger data set also need not prefer a larger ℓ : additional statistics can resolve background structure and move the optimum downward. The upper range is therefore selected from boundary occupancy, likelihood ordering, optimizer stability, and fixed-amplitude

signal-recovery diagnostics, never from a favorable observed limit or local p_0 . These checks can justify a bounded candidate range, but they do not by themselves establish frequentist coverage.

The v4.2 result uses the dataset-specific factors in Table 4. The lower factors retain their earlier full-refit closure support; changing the 2016 upper factor does not requailify those closure studies at full statistics.

Dataset	$k_{\min,d}$	$k_{\max,d}$	v4.2 status
2015 full	1.0	8	unchanged from v4
2016 full	0.9	12	accepted; conditional
2021 10%	1.1	15	unchanged from v4

Table 4: Resolution-scaled RBF length-scale factors in the v4.2 result. The factors multiply $\sigma_x(m)$ in Eq. (24). Only the 2016 upper factor differs from v4. Factor 12 is accepted as the current full-2016 setting, but the observed-data selection remains distinct from a pre-unblinding freeze or direct coverage qualification.

The lower factors are based on full-refit functional-form closure scans, with the observed-spectrum result kept out of the selection criterion. For 2015, the 100-toy scan slightly prefers $k_{\min} = 1.1$ in its aggregate score, but $k_{\min} = 1.0$ has similar pull-width calibration and a weaker expected-reach proxy. The latter is therefore retained as the conservative frozen baseline rather than described as the numerical score winner. For 2016, the 25-toy continuation directly supports the change to $k_{\min} = 0.9$: its nonzero-injection RMS pull mean is 0.321 and its median pull width is 0.948, compared with 0.403 and 0.889 for $k_{\min} = 1.1$. It gives the smallest aggregate calibration score among the scanned 0.9, 1.0, and 1.1 rows and reduces the coherent apparent signal-recovery bias that motivated the follow-up. For 2021, the corrected matched-reference scan finds small rank separations among the same three settings. The $k_{\min} = 1.1$ row has the median pull width closest to unity and the best extraction-level ε^2 proxy, while the 0.9 and 1.0 rows have slightly smaller aggregate scores. The frozen 1.1 choice is therefore a calibration–reach tradeoff, not a claim that it wins every diagnostic. These development studies used the data fractions stated in Section 5; they should not be read as a new full-statistics validation of the v4.1 upper range.

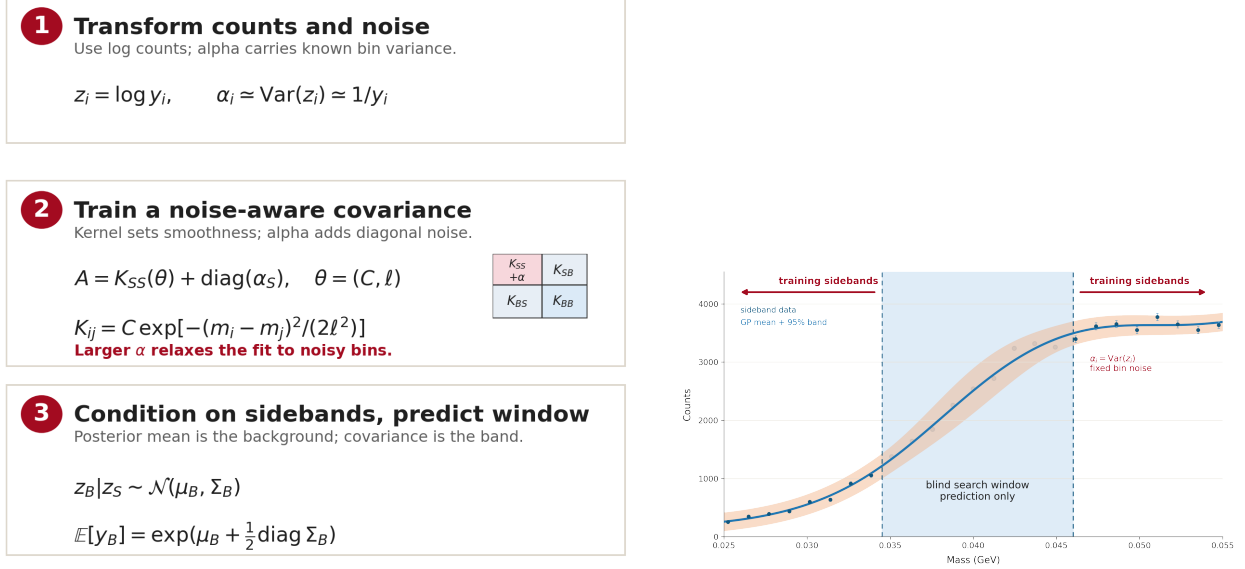
Version 4 used upper factors 8, 8, and 15 for 2015, 2016, and 2021. In its reviewed observed scan, the 2016 optimum reached its upper bound at all 142 hypotheses. The v4.1 range study therefore changes only $k_{\max,2016}$ and evaluates factors 8, 10, 12, 15, and 20 on the same 39–180 MeV mass grid. A fit is counted as upper-bound occupied when $\ell_{\text{opt}}/\ell_{\text{max}} \geq 0.999$. The respective occupancies are 142, 56, 0, 0, and 0 of 142. Factor 12 is the first scanned setting with no occupied hypotheses. It is also followed by a numerical plateau: from factor 12 to 15, the largest pointwise change in the 2016 observed yield upper limit is 0.083%, the largest $|\Delta Z|$ is 0.00255, and the log-marginal-likelihood differences lie between -4.5×10^{-6} and 8.0×10^{-6} . The factor-15 to factor-20 comparison is similarly stable. The selection is based on removal of the artificial boundary and the next-setting plateau, not on which observed limit is tighter or which local p_0 is smaller.

Because this diagnostic was prompted by the v4 observed spectrum, factor 12 is accepted for the v4.2 result with an explicit post-selection qualification; it is not presented as a pre-unblinding choice or a coverage-calibrated nominal. Production-matched functional-form closure and direct coverage remain separate requirements. The configurations continue to retain a dataset-wide median-resolution floor on $\ell_{\max,d}$ so that the ceiling does not become artificially small where the local resolution is exceptionally tight. This is a guard based on a representative σ_x value, not a floor derived from luminosity or event statistics. Alternative upper bounds, fixed-background fits, and

the historical common-lower-bound overlays remain robustness studies collected in the appendices.

An earlier 2021 source/exposure pilot used a nested-Poisson construction. It motivated the later matched-refit studies but is not part of the v4.9 statistical definition.

Figure 23 summarizes the same construction in the language used by the implemented likelihood. The sideband bins set the log-count training sample and its heteroscedastic bin-noise assignment, the kernel hyperparameters are optimized only on those sidebands, and the trained GP is then conditioned to predict the mean and covariance in the blinded mass window. That predicted count-space mean and covariance are the background inputs passed to the local profiled signal extraction.



(a) Log-count transform, noise-aware covariance training, and window prediction. The filled covariance boxes indicate the sideband-sideband, sideband-blind, and blind-blind kernel blocks used when conditioning the GP.

(b) Sideband-only training and blinded-window prediction. Filled markers show binned count data, vertical error bars show the fixed per-bin statistical noise, and the horizontal reference lines are count-axis guides used to compare sideband and blinded-window scales.

Figure 23: Kernel and blinded-window construction used by the GPR workflow. The GP is trained on sideband bins with fixed bin-noise variances, not on the blinded bins. The posterior prediction in the blind window supplies the background mean and covariance that enter the local likelihood; the shaded band in the schematic denotes the GP posterior interval and the dashed vertical lines mark the blinded-window boundaries.

4.7 Blind-window prediction in count space

For each mass point, the package trains the GP only on the bins outside the training exclusion window and predicts the posterior mean and covariance of the latent smooth log-rate in the blinded bins. Since the fit is performed in log-count space, that posterior is mapped back to count space through the standard lognormal moment relations:

$$\mu_i = \exp\left(\mu_i^{(\log)} + \frac{1}{2} \Sigma_{ii}^{(\log)}\right), \quad (28)$$

$$\text{Cov}(\lambda_i, \lambda_j) = \mu_i \mu_j \left[\exp\left(\Sigma_{ij}^{(\log)}\right) - 1 \right]. \quad (29)$$

The resulting blind-window mean vector $\mathbf{b}_d(m)$ and covariance matrix $\mathbf{C}_d(m)$ are the central statistical objects entering both the local extraction fit and the CL_s limit. In other words, once the GP prediction has been translated back into count space, the analysis no longer works with an abstract regression model; it works with an explicit multivariate background prediction for the latent expected counts in the blinded bins. The Poisson counting fluctuation of the observed $n_{d,i}$ is handled separately by the likelihood, so $\mathbf{C}_d$ should be read as the covariance of the GP-predicted background mean rather than the covariance of the observed histogram counts themselves.

To propagate that correlated GP uncertainty into the profile likelihood, the analysis uses the Cholesky factorization

$$\mathbf{C}_d(m) = L_d(m)L_d^\top(m), \quad (30)$$

with $L_d(m)$ the lower-triangular Cholesky factor of the positive-semidefinite blind-window covariance. Introducing an auxiliary standard-normal nuisance vector $\boldsymbol{\theta}_d \sim \mathcal{N}(0, I)$ and the background-deformation field $\Delta\mathbf{b}_d(m) = L_d(m)\boldsymbol{\theta}_d$ then gives $\Delta\mathbf{b}_d \sim \mathcal{N}(0, \mathbf{C}_d)$. This is the standard HEP construction for encoding correlated Gaussian-constrained nuisance variations inside a profile likelihood model [14, 15]. In the present analysis it is especially important because the GP prediction induces nontrivial bin-to-bin correlations across the blinded window; carrying those correlations through $L_d\boldsymbol{\theta}_d$ lets the local significance and CL_s limit respond to coherent background deformations rather than misinterpreting the GP uncertainty as independent per-bin noise.

4.8 Signal model and conversion to ε^2

For each dataset d and mass hypothesis m_0 , the package first constructs a full-range Gaussian line-shape template over the histogram binning,

$$w_{d,i}^{\text{full}}(m_0) = \int_{e_i}^{e_{i+1}} \frac{1}{\sqrt{2\pi}\sigma_d(m_0)} \exp\left[-\frac{(m - m_0)^2}{2\sigma_d(m_0)^2}\right] dm, \quad (31)$$

where e_i and e_{i+1} are the lower and upper edges of histogram bin i . The template is normalized so that

$$\sum_{i \in \text{full range}} w_{d,i}^{\text{full}}(m_0) = 1. \quad (32)$$

The template entering the local likelihood is then the blinded-bin slice of this full-range object,

$$w_{d,i}(m_0) = \begin{cases} w_{d,i}^{\text{full}}(m_0), & i \in \mathcal{B}_d(m_0), \\ \text{not used in the local fit,} & i \notin \mathcal{B}_d(m_0). \end{cases} \quad (33)$$

By the blinded-bin slice we mean exactly this restriction of the full normalized Gaussian template to the subset of bins in $\mathcal{B}_d(m_0)$; the template is not renormalized after the restriction. Its blinded-bin sum is therefore

$$f_{\text{blind},d}(m_0) = \sum_{i \in \mathcal{B}_d(m_0)} w_{d,i}(m_0) < 1, \quad (34)$$

with a value near 0.90 for a $1.64\sigma_d$ half-width and near 0.98 for the current $2.25\sigma_d$ observed-limit half-width. The signal amplitude A_d is defined as the *total* local Gaussian signal yield associated with the mass hypothesis, not merely the yield inside the blind window. The expected signal contribution entering bin i of the local likelihood is therefore

$$s_{d,i}(A_d; m_0) = A_d w_{d,i}(m_0), \quad (35)$$

so the total expected signal actually visible inside the blind window is $f_{\text{blind},d}(m_0) A_d$.

This convention is more physical than blind-window renormalization because it preserves the fact that a finite blind region never contains the entire Gaussian line shape. It also keeps the signal parameter used in the fit aligned with the full-histogram injection model used in the procedural toy studies.

The conversion between signal yield and dark-photon coupling follows the standard fixed-target relation between A' production and radiative trident production [16, 5, 2]. In the present notation,

$$\left. \frac{dN_{A'}}{dm} \right|_{m=m_{A'}} = \frac{3\pi}{2N_{\text{eff}}\alpha_{\text{EM}}} m_{A'} \varepsilon^2 \left. \frac{dN_{\gamma^*}}{dm} \right|_{m=m_{A'}}, \quad (36)$$

where α_{EM} is the fine-structure constant, the factor $3\pi/2$ is the standard narrow-width coefficient relating on-shell A' production to radiative trident production, and N_{eff} counts the number of kinematically open visible decay channels contributing to the total A' width. Below the dimuon threshold, only the e^+e^- channel is open and the minimal-model conversion has $N_{\text{eff}} = 1$. The present GPR implementation keeps this e^+e^- normalization as the analysis convention for the prompt electron-channel yield; minimal dark-photon interpretations above $2m_\mu$ require updating the visible branching factor to include the newly open $\mu^+\mu^-$ channel. Writing the radiative differential rate as

$$\frac{dN_{\gamma^*}}{dm} = f_{\text{rad},d}(m) \rho_d(m), \quad (37)$$

where $\rho_d(m)$ is the differential prompt background rate in the selected invariant-mass spectrum, the package converts between A_d and ε^2 through

$$\varepsilon^2 = \frac{2\alpha_{\text{EM}} A_d}{3\pi m f_{\text{rad},d}(m) \rho_d(m)}, \quad (38)$$

where $\rho_d(m)$ is estimated from the observed histogram as a local counts-per-GeV density in the same mass interval used for the nominal blind window,

$$\rho_d(m) \equiv \frac{N_d[m - n_{\text{blind}}\sigma_d(m), m + n_{\text{blind}}\sigma_d(m)]}{2 n_{\text{blind}}\sigma_d(m)}, \quad (39)$$

with $n_{\text{blind}} = 1.64$ retained as the current prompt-density normalization window even when the extraction blind window is widened to $2.25\sigma_d$. This keeps the yield-to- ε^2 conversion tied to the historically used local density scale while the likelihood itself can recover the wider signal tails. The density-window width remains configurable so that alternative conventions can be reproduced when needed. It is convenient to define the dataset-specific conversion factor

$$K_d(m) \equiv A_d(\varepsilon^2 = 1) = \frac{3\pi m f_{\text{rad},d}(m) \rho_d(m)}{2\alpha_{\text{EM}}}, \quad (40)$$

so that a shared coupling ε^2 predicts a dataset-specific total signal yield $A_d = K_d(m)\varepsilon^2$.

Unless stated otherwise, the methodology section uses the nominal radiative fraction $f_{\text{rad},d}(m)$. The optional penalty that propagates normalization uncertainty into ε^2 is treated as a systematic variation rather than as part of the core GP fit and is discussed separately in Section 3.6. The numerical penalty values are therefore result-configuration choices, not ingredients of the background interpolation or local extraction fit. When that systematic option is enabled for dataset d , the effective radiative fraction is

$$f_{\text{rad},d}^{\text{eff}}(m) = (1 - \delta_{f,d}) f_{\text{rad},d}(m), \quad (41)$$

so the same signal-yield limit maps to a weaker ε^2 limit.

4.9 Statistical inference in a single dataset

For a fixed dataset d and mass hypothesis m , let $\mathbf{n}_d$ denote the vector of observed blinded-bin counts, $\mathbf{b}_d$ the GP-predicted blinded-bin mean, $\mathbf{C}_d = L_d L_d^\top$ the corresponding covariance of Section 4.7 and Eq. (30), and $\mathbf{w}_d(m)$ the non-renormalized blinded-bin slice of the full Gaussian template defined in Eq. (33). The profiled local likelihood is

$$\mathcal{L}_d(A_d, \boldsymbol{\theta}_d) = \left[\prod_{i=1}^{N_d(m)} \text{Pois}(n_{d,i} | b_{d,i} + (L_d \boldsymbol{\theta}_d)_i + A_d w_{d,i}(m)) \right] \exp\left(-\frac{1}{2} \boldsymbol{\theta}_d^\top \boldsymbol{\theta}_d\right), \quad (42)$$

where the Gaussian nuisance vector $\boldsymbol{\theta}_d \sim \mathcal{N}(0, I)$ induces the correlated background-deformation field $L_d \boldsymbol{\theta}_d$ defined below Eq. (30), and $N_d(m)$ is the number of bins in the local blinded set $\mathcal{B}_d(m)$. The signal contribution entering the Poisson mean is exactly the blinded-bin restriction of Eq. (35).

At fixed m , profiling Eq. (42) with no physical sign constraint on A_d yields the unconstrained extraction estimator $\hat{A}_d^{\text{unc}}$ and its local uncertainty $\sigma_{A,d}$. This unconstrained fit is used for closure, linearity, and pull studies because it allows downward background fluctuations to be captured by the estimator itself rather than being clipped at zero. For discovery and production upper limits, however, the parameter of interest is restricted to the physical region $A_d \geq 0$, and the corresponding physical best fit is

$$\hat{A}_d = \max(0, \hat{A}_d^{\text{unc}}). \quad (43)$$

We denote by $\hat{\boldsymbol{\theta}}_d$ the nuisance-parameter maximizer at the unconditional best fit and by $\hat{\boldsymbol{\theta}}_d(A_d)$ the conditional maximum likelihood estimator of $\boldsymbol{\theta}_d$ for fixed A_d .

4.9.1 Local significance

The single-dataset discovery statistic is the bounded profile-likelihood ratio

$$q_{0,d}(m) = -2 \ln \frac{\mathcal{L}_d(A_d = 0, \hat{\boldsymbol{\theta}}_d(0))}{\mathcal{L}_d(\hat{A}_d, \hat{\boldsymbol{\theta}}_d)}, \quad (44)$$

with $\hat{A}_d \geq 0$. The local one-sided discovery p -value and the corresponding Gaussian significance are then obtained with the asymptotic Cowan-et-al. mapping [15]. The resulting local-significance definition is explicitly one-sided and discovery facing: downward fluctuations are allowed in the unconstrained fit used for closure studies, but they do not generate a positive local discovery significance. This profiled multi-bin test replaces the older sum-count approximation used in the earliest drafts and is the only local-significance definition treated as current in this note.

4.9.2 CL_s upper limits

To test a fixed non-negative signal amplitude A_d , the production limit setting uses the bounded one-sided profile-likelihood statistic

$$\tilde{q}_{A_d}(m) = \begin{cases} 0, & \hat{A}_d^{\text{unc}} > A_d, \\ -2 \ln \frac{\mathcal{L}_d(A_d, \hat{\theta}_d(A_d))}{\mathcal{L}_d(\hat{A}_d^{\text{unc}}, \hat{\theta}_d)}, & 0 \leq \hat{A}_d^{\text{unc}} \leq A_d, \\ -2 \ln \frac{\mathcal{L}_d(A_d, \hat{\theta}_d(A_d))}{\mathcal{L}_d(0, \hat{\theta}_d(0))}, & \hat{A}_d^{\text{unc}} < 0, \end{cases} \quad (45)$$

which is the standard bounded Cowan construction for upper limits [15]. The corresponding modified-frequentist tail areas are

$$\text{CL}_{s+b}(A_d; m) = P(\tilde{q}_{A_d} \geq \tilde{q}_{A_d}^{\text{obs}} | A_d + \text{bkg}), \quad \text{CL}_b(A_d; m) = P(\tilde{q}_{A_d} \geq \tilde{q}_{A_d}^{\text{obs}} | \text{bkg}), \quad (46)$$

and

$$\text{CL}_s(A_d; m) = \frac{\text{CL}_{s+b}(A_d; m)}{\text{CL}_b(A_d; m)}. \quad (47)$$

Dividing by CL_b prevents a downward background fluctuation from producing an artificially aggressive exclusion in a region with little intrinsic sensitivity [17, 18]. At confidence level $1 - \alpha$, the single-dataset upper limit is

$$A_{\text{UL},d}(m; \alpha) = \min\{A_d : \text{CL}_s(A_d; m) \leq \alpha\}. \quad (48)$$

The v4 result uses $\alpha = 0.10$, corresponding to a 90% CL upper limit; retained historical 95% studies use $\alpha = 0.05$. The package reports the limit both in signal-yield space and, after applying the dataset-specific conversion factor of Eq. (40), in ε^2 space.

4.10 Asymptotic and toy calibration

The local-significance mapping introduced above is treated asymptotically throughout the production scan. For upper limits, the same bounded statistic $\tilde{q}_{A_d}$ is calibrated in one of two ways. In asymptotic mode, the background-only Asimov data set is used to evaluate the reference quantity $\tilde{q}_{A_d,A}$, so that

$$\text{CL}_{s+b}(A_d; m) = 1 - \Phi(\sqrt{\tilde{q}_{A_d}^{\text{obs}}}), \quad \text{CL}_b(A_d; m) = \Phi(\sqrt{\tilde{q}_{A_d,A}} - \sqrt{\tilde{q}_{A_d}^{\text{obs}}}). \quad (49)$$

In toy mode, the same statistic $\tilde{q}_{A_d}$ is calibrated directly with background-only and signal-plus-background pseudoexperiments rather than by switching to a different ordering variable. The shared-coupling combination described below uses the identical logic after replacing the single-dataset amplitude A_d with the common parameter of interest ε^2 .

The pseudoexperiment program remains important even with the asymptotic formulae in place because the HPS likelihood combines a multi-bin Poisson term, a GP-conditioned blind-window covariance, a non-renormalized Gaussian signal template, and a mass-dependent yield-to- ε^2 conversion. The pull, spurious-signal, and injection-recovery diagnostics of Section 5 test this machinery under the declared conditional backgrounds. They are not direct CL_s coverage studies or a scan-wide calibration of Z .

4.11 Matched-reference injection–extraction diagnostics

The v4.5 contract defines an injection axis through a background-only reference fit that uses the same support, binning, exclusion mask, kernel bounds, preprocessing, and GP-refit path as the corresponding toy extraction. If $\mathcal{F}[\mathbf{n}; \mathbf{w}]$ denotes that full procedure for counts $\mathbf{n}$ and the blind-window slice of the unit-normalized full-range signal template $\mathbf{w}$, then

$$(\hat{A}_{\text{ref}}, \sigma_{A,\text{ref}}) = \mathcal{F}[\mathbf{n}_{\text{ref}}; \mathbf{w}], \quad A_{\text{inj}}(z) = z \sigma_{A,\text{ref}}, \quad z \in \{0, 1, 3, 5\}. \quad (50)$$

The reference uncertainty is a fixed calibration of the injection coordinate at a given source, toy index, and mass. It is not the post-injection uncertainty returned for an individual extraction.

For accepted state t , the signed pull and paired response are

$$p_t(z) = \frac{\hat{A}_t(z) - A_{\text{inj},t}(z)}{\sigma_{A,t}(z)}, \quad (51)$$

$$R_t(z) = \frac{\hat{A}_t(z) - \hat{A}_t(0)}{A_{\text{inj},t}(z)}, \quad z > 0. \quad (52)$$

The zero-injection mean pull diagnoses a conditional spurious-signal offset under the declared generating truth. Its width probes the returned uncertainty scale, while R_t separates injection response from a background-specific zero-signal offset. These quantities use the toy-level $\sigma_{A,t}$ in the denominator. A finite 25-background sample supports two-sided 90% Student- t intervals on the mean and descriptive widths, not a direct coverage statement.

Full refitting and no-refit displays answer different questions. Holding a GP state fixed isolates the local profile-likelihood extraction. Refitting the GP on every pseudo-spectrum additionally tests background adaptation and possible signal absorption. Version 4.9 uses the latter matched-refit path. Optimizer acceptance is decided from fit state, likelihood, reproducibility, covariance, and kernel-coordinate diagnostics before amplitudes or pulls are summarized; no row is selected because its pull is favorable.

4.12 Observed-limit toy diagnostics

The analysis uses two distinct families of p -values for different purposes. The first is the discovery statistic of Eq. (44), summarized through $q_0(m)$, $p_0(m)$, and $Z_{\text{local}}(m)$. The second consists of diagnostics derived from the background-only CL_s limit ensemble at each mass. For N_{toys} background-only pseudoexperiments and an observed limit $\varepsilon_{\text{UL,obs}}^2(m)$ at the confidence level under study, those quantities are

$$p_{\text{strong}}(m) = \frac{1}{N_{\text{toys}}} \sum_{t=1}^{N_{\text{toys}}} \mathbf{1} \left[\varepsilon_{\text{UL},t}^2(m) \leq \varepsilon_{\text{UL,obs}}^2(m) \right], \quad (53)$$

$$p_{\text{weak}}(m) = \frac{1}{N_{\text{toys}}} \sum_{t=1}^{N_{\text{toys}}} \mathbf{1} \left[\varepsilon_{\text{UL},t}^2(m) \geq \varepsilon_{\text{UL,obs}}^2(m) \right], \quad (54)$$

$$p_{\text{two}}(m) = \min[1, 2 \min(p_{\text{strong}}(m), p_{\text{weak}}(m))]. \quad (55)$$

Here p_{strong} quantifies how unusually *strong* the observed limit is relative to the background-only toy ensemble, while p_{weak} quantifies how unusually *weak* it is. The two-sided quantity p_{two} is retained only as a symmetric diagnostic; because it responds to both unusually strong and unusually weak limits, it is less targeted than the one-sided discovery or exclusion tests and is not used as a search

statistic. Figure 24 gives the compact fixed-mass definition of these quantities, while Figure 25 shows representative strong-limit, weak-limit, and bounded median-case examples from the same toy-limit logic.

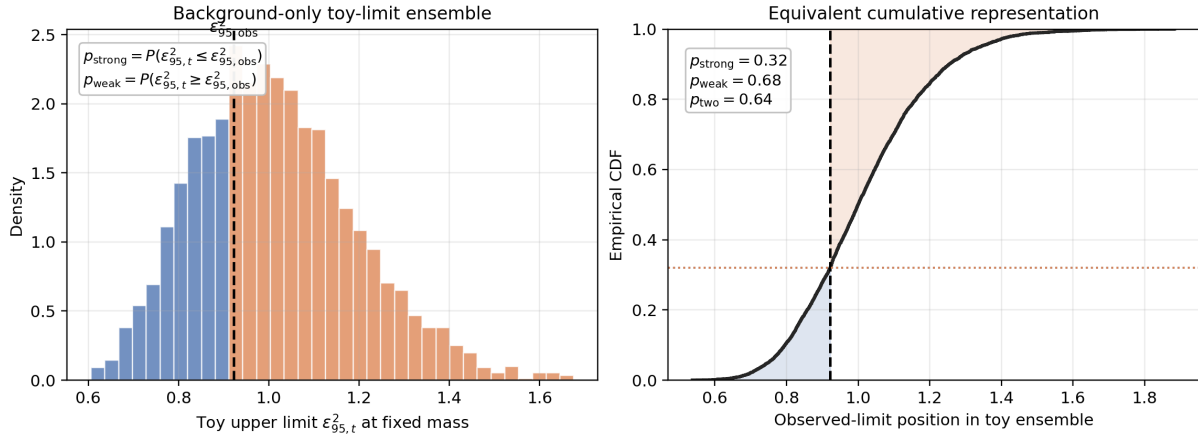


Figure 24: Illustrative toy-limit diagnostic at fixed mass. The left panel shows a background-only ensemble of observed CL_s upper limits, and the right panel shows the corresponding empirical cumulative distribution. The vertical line marks the observed limit. The lower-tail area defines p_{strong} , the upper-tail area defines p_{weak} , and $p_{two} = \min[1, 2 \min(p_{strong}, p_{weak})]$ is used only as a symmetric diagnostic summary.

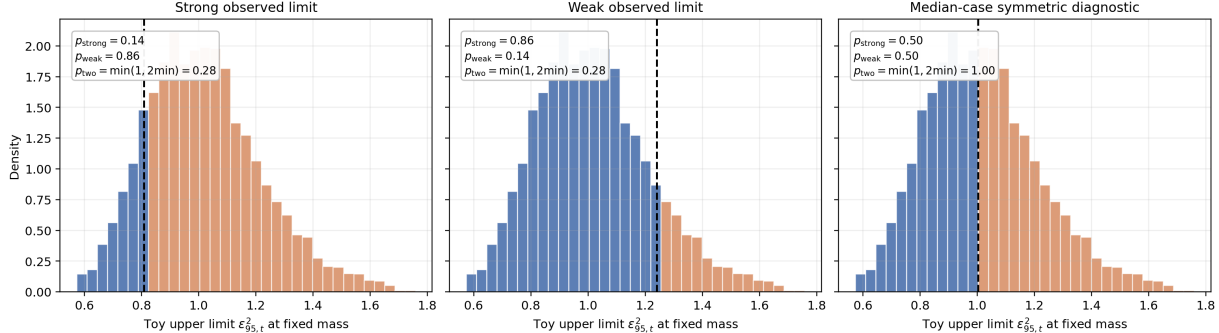


Figure 25: Representative fixed-mass observed-limit toy diagnostics. The left panel shows an unusually strong observed limit relative to the background-only toy ensemble, the middle panel shows an unusually weak observed limit, and the right panel shows a near-median case where the bounded symmetric summary saturates at $p_{two} = 1$. These are limit diagnostics, not discovery statistics.

These quantities are therefore limit diagnostics rather than discovery-significance measures in the sense of p_0 . If the observed limit lies near the median of the background-only ensemble, then roughly half of the toys lie above it and half lie below it, making both p_{strong} and p_{weak} naturally $\mathcal{O}(0.5)$. Their finite resolution is $1/N_{toys}$. In the v4.2 combined ensemble, $N_{toys} = 300$, so a one-count change is $1/300 \simeq 0.0033$ and a zero-count tail is reported as 0 of 300 rather than as an exact zero probability. These toy-limit diagnostics cannot by themselves establish or refute a high-significance local excess in the discovery statistic defined by the one-sided local p_0 mapping above.

4.13 Shared- ε^2 combination across datasets

At a common mass hypothesis m , each active dataset contributes a distinct blinded-bin count vector, GP background prediction, covariance matrix, and Gaussian signal template. Let the set of datasets that cover the hypothesis be $\mathcal{D}(m)$. For each $d \in \mathcal{D}(m)$, define

$$\mathbf{n}_d(m), \quad \mathbf{b}_d(m), \quad \mathbf{C}_d(m) = L_d L_d^\top, \quad \mathbf{w}_d(m), \quad K_d(m), \quad (56)$$

where $\mathbf{w}_d(m)$ is the blinded-bin template slice of Eq. (33), $K_d(m)$ is the yield-per-unit- ε^2 conversion of Eq. (40), and L_d is the Cholesky factor of the GP covariance in that dataset. When the radiative-fraction penalty of Eq. (10) is enabled, the conversion becomes

$$K_d^{\text{eff}}(m) = (1 - \delta_{f,d}) K_d(m), \quad (57)$$

so the expected signal vector per unit ε^2 is reduced coherently in every active dataset. The per-dataset likelihood is then

$$\mathcal{L}_d(\varepsilon^2, \boldsymbol{\theta}_d) = \left[\prod_{i=1}^{N_d(m)} \text{Pois}(n_{d,i} \mid b_{d,i} + (L_d \boldsymbol{\theta}_d)_i + \varepsilon^2 K_d^{\text{eff}}(m) w_{d,i}(m)) \right] \exp\left(-\frac{1}{2} \boldsymbol{\theta}_d^\top \boldsymbol{\theta}_d\right). \quad (58)$$

The baseline combined likelihood is the product over datasets,

$$\mathcal{L}_{\text{comb}}(\varepsilon^2, \{\boldsymbol{\theta}_d\}) = \prod_{d \in \mathcal{D}(m)} \mathcal{L}_d(\varepsilon^2, \boldsymbol{\theta}_d). \quad (59)$$

Equivalently, one may concatenate all blinded-bin vectors into a single object. In the following equations, $\parallel$ denotes vector concatenation, e.g. $\mathbf{n}_1 \parallel \mathbf{n}_2$:

$$\mathbf{n} = \mathbf{n}_1 \parallel \mathbf{n}_2 \parallel \cdots, \quad (60)$$

$$\mathbf{b} = \mathbf{b}_1 \parallel \mathbf{b}_2 \parallel \cdots, \quad (61)$$

$$\mathbf{s}_{\text{unit}} = (K_1^{\text{eff}} \mathbf{w}_1) \parallel (K_2^{\text{eff}} \mathbf{w}_2) \parallel \cdots, \quad (62)$$

$$\boldsymbol{\theta} = \boldsymbol{\theta}_1 \parallel \boldsymbol{\theta}_2 \parallel \cdots, \quad (63)$$

$$\mathbf{C}_{\text{comb}} = \text{blockdiag}(\mathbf{C}_1, \mathbf{C}_2, \dots), \quad L_{\text{comb}} = \text{blockdiag}(L_1, L_2, \dots), \quad (64)$$

so that the total expected mean is

$$\boldsymbol{\lambda}(\varepsilon^2, \boldsymbol{\theta}) = \mathbf{b} + L_{\text{comb}} \boldsymbol{\theta} + \varepsilon^2 \mathbf{s}_{\text{unit}}, \quad \mathbf{C}_{\text{comb}} = L_{\text{comb}} L_{\text{comb}}^\top. \quad (65)$$

The bounded upper-limit statistic for the shared coupling, $\tilde{q}_{\varepsilon^2}(m)$, is defined exactly as in Eq. (45) after replacing the single-dataset amplitude with the common parameter of interest ε^2 and the single-dataset likelihood with $\mathcal{L}_{\text{comb}}$. Two points are essential. First, the baseline combination shares only the parameter of interest ε^2 . The nuisance vectors $\boldsymbol{\theta}_d$ are independent between datasets, and no common radiative-fraction, luminosity, or resolution nuisance parameters are profiled across runs in the current production model. Second, the combination is performed at the likelihood level, not by averaging precomputed upper-limit curves. At fixed ε^2 ,

$$-2 \ln \lambda_{\text{comb}}(\varepsilon^2) = \sum_{d \in \mathcal{D}(m)} -2 \ln \lambda_d(\varepsilon^2), \quad (66)$$

after profiling the independent nuisance blocks. The sensitivity gain therefore comes from the additive information content of statistically independent channels, each with its own detector response

and normalization factor, rather than from any post hoc merging of already-compressed outputs. This is the standard common-POI structure used in multi-channel HEP combinations [14, 19]. The combined upper limit is set directly on the shared coupling,

$$\varepsilon_{\text{UL}}^2(m; \alpha) = \min\{\varepsilon^2 : \text{CL}_s(\varepsilon^2) \leq \alpha\}, \quad (67)$$

with $\alpha = 0.10$ for the v4 90% CL result. The combined local significance is evaluated from the same profiled likelihood with the common signal vector $\varepsilon^2 \mathbf{s}_{\text{unit}}$. Inverse-variance combinations are useful diagnostic proxies; they do not define the combined inference.

4.14 Global significance and the look-elsewhere treatment

The physics-facing local significance is the minimum one-sided p_0 in the scan,

$$p_{0,\min}^{\text{obs}} = \min_m p_0^{\text{obs}}(m), \quad q_{0,\max}^{\text{obs}} = \max_m q_0^{\text{obs}}(m). \quad (68)$$

The cleanest global-significance definition is then scan based:

$$p_{\text{global}}^{\text{toy}} = \frac{1}{N_{\text{toys}}} \sum_{t=1}^{N_{\text{toys}}} \mathbf{1}[q_{0,\max}^{(t)} \geq q_{0,\max}^{\text{obs}}], \quad (69)$$

with

$$Z_{\text{global}}^{\text{toy}} = \Phi^{-1}(1 - p_{\text{global}}^{\text{toy}}). \quad (70)$$

When a dense full-scan toy calibration is not yet frozen, the workflow monitors the look-elsewhere effect through an effective-trials approximation based on the scan span and detector resolution [20]. In that representation,

$$p_{\text{global}}^{\text{Sidak}} = 1 - \left(1 - p_{0,\min}^{\text{obs}}\right)^{N_{\text{eff}}}, \quad Z_{\text{global}}^{\text{Sidak}} = \Phi^{-1}(1 - p_{\text{global}}^{\text{Sidak}}). \quad (71)$$

For small $p_{0,\min}^{\text{obs}}$, Eq. (71) reduces to the familiar $p_{\text{global}} \simeq N_{\text{eff}} p_{0,\min}$ form.

For the present result, Eq. (71) supplies a separate analytic look-elsewhere reference for the local scan. The effective number of trials is evaluated from the stated scan range and the mass-dependent resolution, so the approximation accounts for the strong correlation between adjacent 1 MeV-spaced hypotheses rather than treating every scan point as independent. It can be compared with the local p -value scan, but it is not a toy-calibrated global discovery probability. A background-only ensemble of the frozen search maximum, $q_{0,\max}$, remains the required calibration for any future discovery-level claim; when that ensemble is available, Eq. (69) supersedes the analytic approximation.

4.15 Signal absorption and alternative background-profile treatments

The production baseline documented here is the additive Gaussian-profile treatment of blind-window background uncertainty. That choice is natural because the GP delivers an additive mean vector and covariance matrix in the blinded bins, and the entire current toy-validation program is built around that object. At the same time, the remaining signal-absorption question should be stated explicitly. The newly adopted leakage-retaining signal model removes one obvious source of bias associated with blind-window renormalization, but it does not by itself prove that the full procedural toy workflow is free of absorption once the GP is refit toy by toy.

The most likely residual sources are the refitting freedom of the GP hyperparameters, especially the effective length scale, and the flexibility of the Gaussian-profile background nuisance itself. For

that reason, the current note keeps the Gaussian-profile model as the mainline baseline while reserving fixed-background and alternative Poisson-style treatments as targeted robustness studies. Those alternatives are not presented as equally mature production choices. They become necessary follow-up tests if production-faithful pseudoexperiments demonstrate that the current nuisance model absorbs an unacceptable fraction of injected signal.

4.16 Dimuon-threshold interpretation

The yield-to-coupling conversion in Eq. (38) is an electron-channel normalization: the fitted resonance amplitude is the $A' \rightarrow e^+e^-$ yield visible in the prompt invariant-mass spectrum. For the minimal kinetically mixed dark photon, this is also the full visible branching interpretation below the dimuon threshold, $m_{A'} < 2m_\mu = 211.3167 \text{ MeV}$, where the $\mu^+\mu^-$ final state is kinematically closed. This is the same fixed-target convention used in the published HPS prompt searches [16, 5, 2]. Above threshold, the same electron-channel yield corresponds to a weaker minimal-model limit on ε^2 because the total width includes the newly open $A' \rightarrow \mu^+\mu^-$ mode [21, 22].

Neglecting the electron mass and remaining below the two-pion hadronic threshold, the charged-lepton width ratio is

$$R_{\mu/e}(m_{A'}) \equiv \frac{\Gamma(A' \rightarrow \mu^+\mu^-)}{\Gamma(A' \rightarrow e^+e^-)} = \sqrt{1 - \frac{4m_\mu^2}{m_{A'}^2}} \left(1 + \frac{2m_\mu^2}{m_{A'}^2} \right), \quad m_{A'} > 2m_\mu, \quad (72)$$

with $R_{\mu/e} = 0$ below threshold. The inverse electron branching fraction used for the minimal-model reinterpretation is therefore

$$N_{\text{eff}}^{\text{BR}}(m_{A'}) \equiv \frac{1}{\mathcal{B}(A' \rightarrow e^+e^-)} = 1 + R_{\mu/e}(m_{A'}), \quad (73)$$

and an electron-channel coupling limit is mapped to the minimal visible-dark-photon convention by

$$\varepsilon_{\text{min}}^2(m_{A'}) = N_{\text{eff}}^{\text{BR}}(m_{A'}) \varepsilon_{ee}^2(m_{A'}). \quad (74)$$

This operation does not change the local fit, the signal-yield upper limit, the CL_s value, or the local p_0 scan; it changes only the coupling interpretation of an electron-channel result. Figure 26 shows the factor applied in the v4 90% CL result obtained with CL_s . The factor is unity below $2m_\mu$, rises continuously above threshold, and reaches $N_{\text{eff}}^{\text{BR}} = 1.725$ at 250 MeV ($\mathcal{B}_{ee} = 0.580$).

The scan range used for a look-elsewhere statement must match the range of the quoted physics claim [20]. If the minimal-dark-photon exclusion is quoted only below 210 MeV, the final scan-level toy or effective-trials calibration should use that same below-threshold range. If the high-mass electron-channel scan is promoted to a 170 MeV to 250 MeV minimal-model result, the corresponding look-elsewhere calibration should include the full extended range as well.

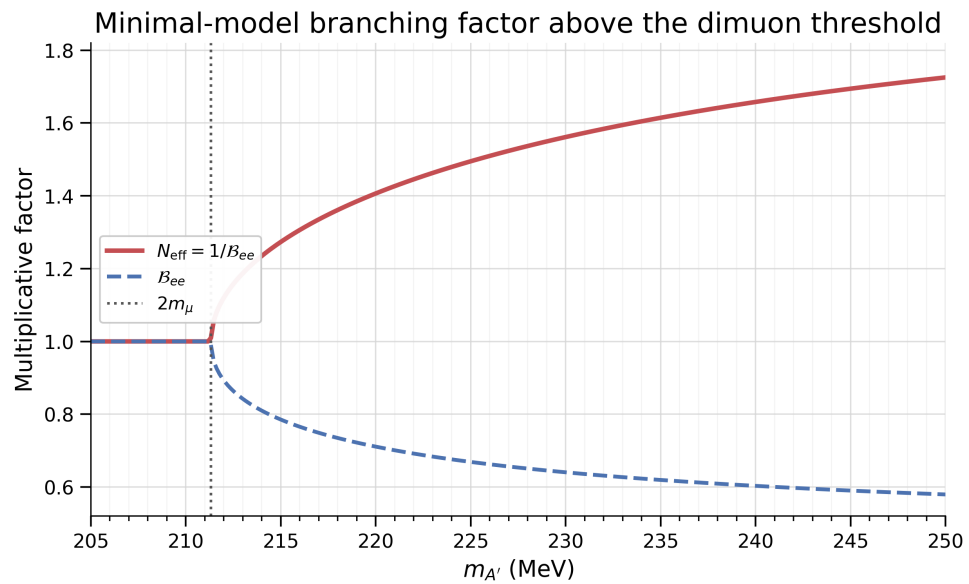


Figure 26: Minimal-model branching factor used to reinterpret electron-channel ε^2 limits above the dimuon threshold. The red curve shows $N_{\text{eff}}^{\text{BR}} = 1/\mathcal{B}(A' \rightarrow e^+e^-)$ and the blue dashed curve shows $\mathcal{B}(A' \rightarrow e^+e^-)$.

5 Background-model validation with full-refit pseudoexperiments

5.1 Validation question and scope

The validation begins with a concrete local question. If a smooth, data-like spectrum contains no resonance, does the complete GPR sideband fit return a spurious signal? If a known signal is added, does the same procedure recover it with an uncertainty of the right scale? The pseudoexperiments exercise the count-space prediction, the mass-dependent blind and training masks, the full-range Gaussian signal template, the matched-reference injection convention, and a fresh GP optimization for every pseudo-spectrum. The statistical definitions are given in Section 4.11; this section presents the physical motivation, the generating spectra, the ensembles, the numerical safeguards, and the closure evidence.

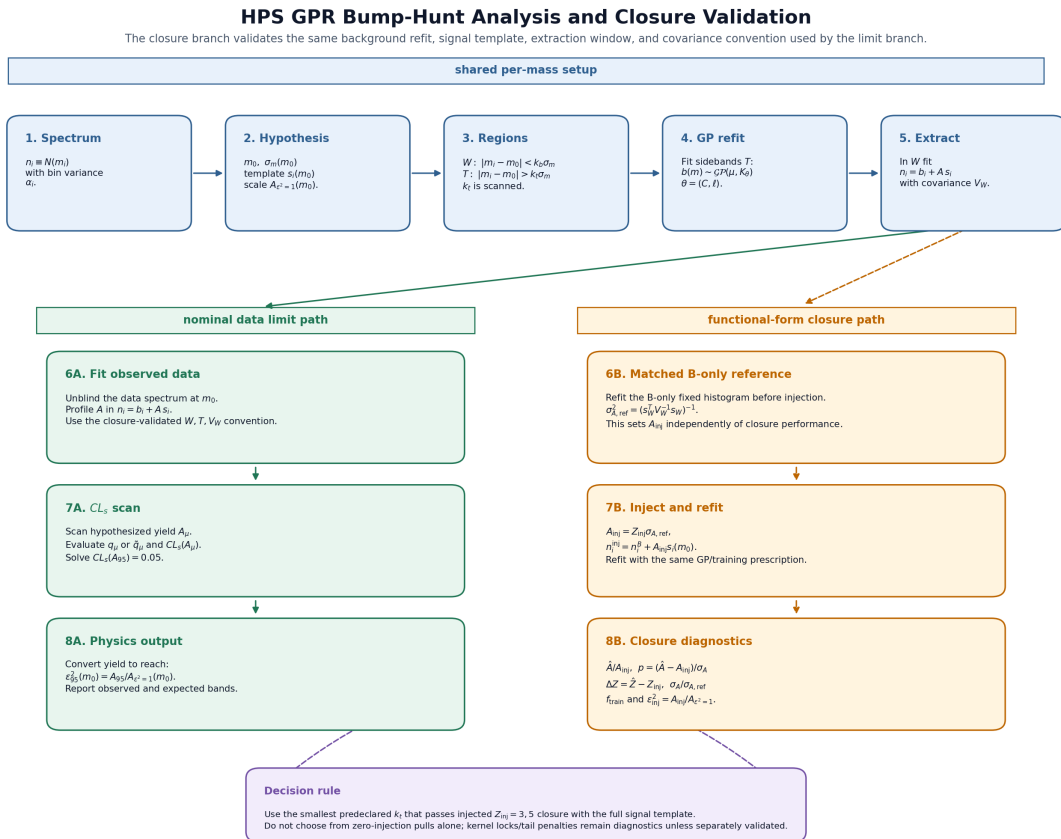


Figure 27: Analysis and validation flow. In the data path, sidebands around each mass hypothesis train the GP and the blinded bins enter the signal-plus-background likelihood. In the validation path, a smooth source-fitted mean is Poisson fluctuated, a known signal may be added, and the complete GP fit and extraction are repeated. The zero-injection ensemble tests spurious signal; nonzero injections test signal recovery and uncertainty calibration. These are extraction-closure tests, not confidence-limit coverage or a scan-wide significance calibration.

Three distinctions are important throughout this section. First, a generating function is a smooth stress truth fitted to an HPS source spectrum; it is not claimed to be the unique physical

parameterization of the trident continuum. Second, closure is conditional on that specified truth and on the analysis card used for the extraction. It therefore tests the behavior of the procedure for data-like spectra without measuring a bias in the observed data. Third, the four exposure lanes are not interchangeable. The $1\% \times 10$ and $1\% \times 100$ spectra derive from the native 1% source, while native 10% and $10\% \times 10$ derive from an independently selected 10% source. Cross-source differences include source and selection shape as well as normalization.

5.2 Closure observables and decision rules

At zero injection, the signed pull $p = \hat{A}/\hat{\sigma}_A$ is a spurious-signal statistic. The ensemble mean tests whether the extraction is centered and the sample standard deviation tests whether the returned uncertainty has the appropriate scale. At nonzero injection, the pull of Eq. (51) tests centering about the known amplitude and the paired response of Eq. (52) separates recovery of the added signal from a background-specific zero-signal offset. Strengths $z = 0, 1, 3, 5$ reuse the same background pseudo-spectrum and are consequently correlated; they are one response curve, not four independent confirmations.

For each 100-background cell we report the usable count n , the pull mean and its two-sided 90% Student- t interval, and the pull width with its exact normal-theory 90% chi-square interval. A cell is reported only if at least 95 of its 100 fits pass the numerical criteria. We account for multiplicity in two ways: a Holm correction over the two planned replacement points at $z = 0$, and a separate Holm correction over the twenty displayed four-lane, five-mass points at $z = 0$ after substitution. The correlated nonzero injections are response diagnostics and do not enter either family. We flag a mean only when its Holm-adjusted two-sided p value is below 0.05 and $|\bar{p}| \geq 0.2$. The absence of such a flag is not proof that the bias is zero.

A practical centering tolerance, $|\bar{p}| < 0.5$, was chosen after the 100-background results were available. It is therefore a descriptive criterion, not a predeclared equivalence test. We report it alongside the t /Holm results, which are more sensitive to small but statistically resolved offsets. For the two replacement points we also ask whether the full 90% interval on the mean lies inside the practical band. A pull width is called consistent with one only when its 90% interval contains one. None of these tests measures CL_s coverage.

5.3 Training exclusion and blind-window choice

The GP must not be trained on the core or near tails of the signal it is later asked to test. A Gaussian signal retains non-negligible weight outside a $1.64\sigma_m$ central interval, so training immediately beside that interval can offer signal-like counts to the background fit and reduce the recovered amplitude. The training-exclusion study therefore varied the sideband boundary before any observed limit was used to choose it.

The widened $2.25\sigma_m$ interval was then used for both the training exclusion and the extraction likelihood. Compared with a $1.64\sigma_m$ extraction plus a separate 1.64 – $2.25\sigma_m$ guard band, this construction gives comparable closure while retaining the near signal tails in the likelihood. The production outcome is a single, explicit mass-dependent geometry for the background and signal fits.

5.4 Generating spectra and the role of functional forms

The principal background toys are generated independently of the GP used for their extraction. A positive analytic intensity is fitted to a native HPS mass spectrum, normalized on the specified toy support, integrated over every histogram bin, and Poisson fluctuated. A complete signal template is

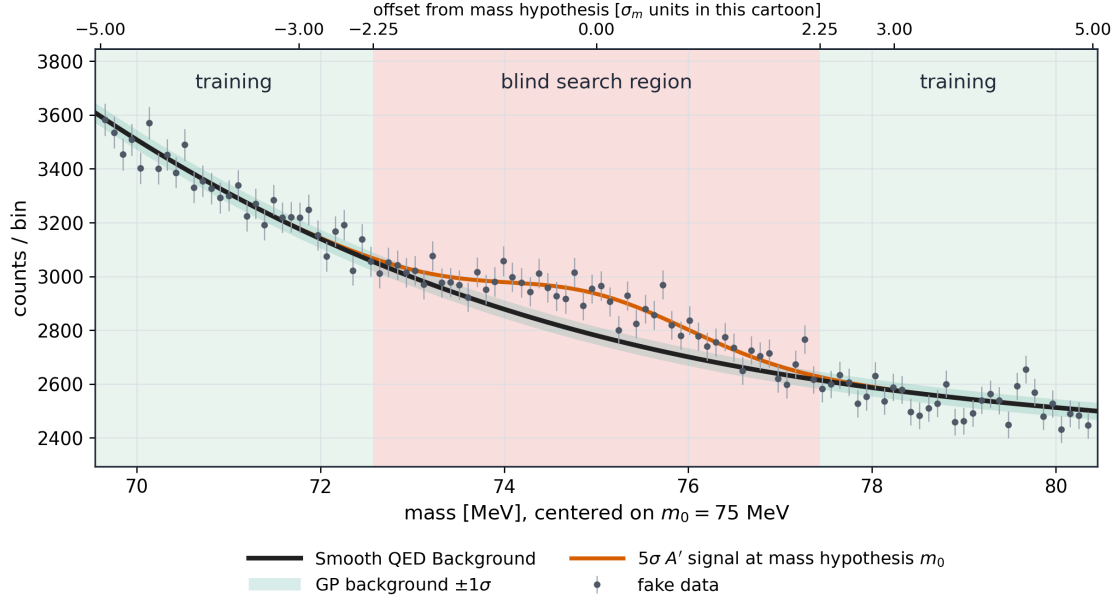


Figure 28: Selected $2.25\sigma_m$ blind and training-exclusion geometry. The GP is trained outside the widened interval, while all bins inside it enter the local signal-plus-background likelihood. This retains the signal tails for extraction without allowing those same bins to influence the sideband fit.

k_{train}	median expected ε_{95}^2 proxy	$\langle \hat{A}/A_{\text{inj}} \rangle_{z=3}$	$\langle \hat{A}/A_{\text{inj}} \rangle_{z=5}$	$\max f_{\text{train}}$
1.96	7.92×10^{-5}	0.730	0.715	0.0530
2.25	9.53×10^{-5}	0.961	0.943	0.0258
2.58	1.08×10^{-4}	1.052	1.036	0.0106
3.00	1.19×10^{-4}	1.049	1.037	0.00288

Table 5: Evidence used to choose the training exclusion. The train-2.25 row substantially reduces the fraction of the signal template available to the GP while recovering the injected amplitude and retaining useful local precision. The first column is a Gaussian-equivalent comparison proxy, not the final CLs limit. The complete note retains the mass-resolved reach, leakage, recovery, and pull batteries.

then added when $z > 0$. Every pseudo-spectrum is subsequently analyzed with a new GP sideband fit. This analytic truth/full-refit construction is the central background-model test because success is not guaranteed by generating toys from the GP model that will later fit them.

The earlier 2015 and 2016 studies used independently fitted shifted sigmoid–power–exponential families. The initial 2021 catalogue selected an unshifted sigmoid–power–exponential model (internal label fSigPowExpQ) for the native 1% source. The later four-lane study used a generalized-gamma threshold model (internal label fGenGammaThresh), fitted independently to the native 1% and native 10% sources. These choices span smooth threshold, power-law, and broad-tail curvature without adding resonance-scale structure. Figure 29 shows the source spectra, selected means, and Poisson-ensemble construction.

The mass-resolved injection scans selected the lower RBF length-scale factors without using the observed limit curve. For the frozen settings, the aggregate zero-signal mean pulls are +0.029, −0.016, and −0.005 for the 2015, 2016 10%, and 2021 1% studies, respectively; the corresponding nonzero-injection signed averages are −0.223, −0.262, and −0.034. Mass-resolved mean, width, significance-residual, and recovery checks support these summaries. They motivate the chosen lower factors but do not replace direct mass-by-mass closure.

5.5 Four-lane 2021 full-refit ensemble

The full-100 2021 validation uses four exposure lanes and five mass hypotheses. The historical baseline is the frozen v4.2/v4.5 matched-refit contract: 40–300 MeV GP support, 50–250 MeV search range, $k_{\min} = 1.1$, $k_{\max} = 15$, five-bin local rebinning, log-mass/log-count preprocessing, the $2.25\sigma_m$ blind and training exclusions, two required sidebands, and twelve randomized optimizer restarts in addition to the nominal start. The original four lanes use the common generalized-gamma threshold model fitted independently to the native 1% and native 10% sources. Their source Pearson χ^2/ndf values are 1.108 and 2.781, respectively. The latter misdescription is largest through the steep threshold turn-on and motivates a separately qualified 65 MeV replacement rather than a post-hoc change to all other masses.

5.6 Targeted validation of the rising threshold

1% $\times$ 10 threshold-turn-on truth. The first replacement is the source-fitted model developed in the dedicated near-threshold study. Over 35–80 MeV it is a logistic turn-on multiplied by an exponentiated degree-five Chebyshev expansion and is joined smoothly to the specified high-mass tail. The degree is selected before any GP extraction: the native-bin deviance per degree of freedom is 0.994 for the native 1% source and the five-bin value is 0.850, with no parameter-bound contact. Its disclosed 80–300 MeV tail Poisson deviance per degree of freedom is 1.122, so this selected 1%-source truth is broadly data-like over the complete 40–300 MeV support rather than only in the fitted turn-on interval. The earlier 20-background result at 65 MeV had mean pull 0.300 and width 1.109; version 4.9.1 replaces that pilot with a predeclared 100-background continuation and reports 90% intervals.

Native-10% 30–300 MeV support. The second replacement models the measured 30–40 MeV shoulder and analyzes it with the complete 30–300 MeV GP support. Its threshold component is a logistic turn-on times a degree-six exponentiated Chebyshev form fitted over 30–80 MeV, joined with a C^2 taper to the frozen smooth anchor. In the native 10% source, the local native-bin and five-bin deviances per degree of freedom are 1.140 and 1.421. Thus the generating spectrum is data-like in the threshold-sensitive region even though the simple historical family is not. Its

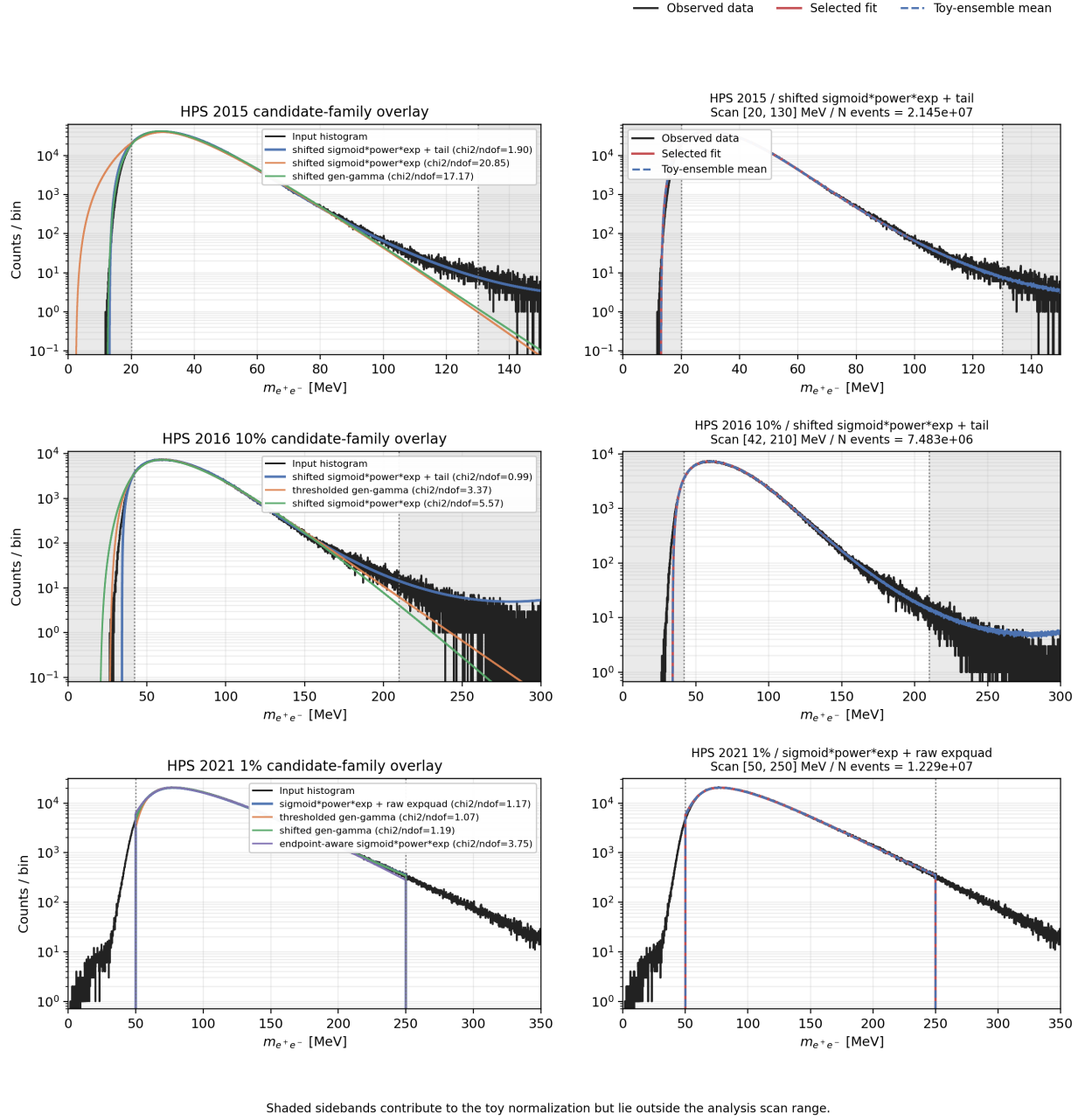


Figure 29: Analytic generating means used in the year-matched functional-form studies. Each dataset row compares the source spectrum with the candidate families and then shows the selected mean against the generated toy ensemble. The functions define smooth positive pseudo-data means and are not asserted to be unique physical models of the continuum.

Displayed cell	Generating mean	GP support	Reason for scheme
$1\% \times 10$, $m \neq 65$ MeV	generalized-gamma threshold model	40–300 MeV	historical 100-background functional-form suite
$1\% \times 10$, 65 MeV	degree-five threshold-turn-on continuation	40–300 MeV	dedicated near-threshold description extended to 100 backgrounds
$1\% \times 100$, all masses	generalized-gamma threshold model	40–300 MeV	historical 100-background functional-form suite
native 10%, $m \neq 65$ MeV	generalized-gamma threshold model	40–300 MeV	historical 100-background functional-form suite
native 10%, 65 MeV	threshold-focused sigmoid–power– exponential mean	30–300 MeV	includes the measured 30–40 MeV turn-on omitted by the historical support
$10\% \times 10$, all masses	generalized-gamma threshold model	40–300 MeV	historical 100-background functional-form suite

Table 6: Composition of the 2021 background-validation display. The two special 65 MeV cells replace the corresponding historical cells; they are never pooled with them. Distinct markers in Figs. 30 and 32 identify the change of generating truth and, for native 10%, the change of GP support.

85–300 MeV tail deviance per degree of freedom is 6.220; this native-10% lane therefore relies on the explicit weak-correlation and tail-influence qualification below rather than on a support-wide goodness-of-fit claim.

This study tests the 65 MeV extraction under the 30–300 MeV support considered in v4.9.1; it does not determine the 36–300 MeV support adopted in v4.9.5. The question here is whether a data-like generating mean that passes the local source-fit tests can be processed by the complete extraction without material signal bias. With log preprocessing, the direct RBF correlation between masses m and m' is

$$\rho(m, m') = \exp \left[-\frac{\log^2(m'/m)}{2\ell^2} \right]. \quad (75)$$

At 65 MeV, the accepted optimized length scales in the two actual replacement ensembles span 0.2784–0.3185 for $1\% \times 10$ and 0.2468–0.2691 for native 10%, with medians 0.2963 and 0.2592. The corresponding direct correlations with 150 MeV span 1.10–3.19% and 0.321–0.800%, respectively. Across the complete 55–70 MeV set the largest direct correlations with 150 MeV occur at 70 MeV and are 5.71% and 1.81%. The correlation decreases rapidly above 150 MeV. These results quantify why a threshold-focused source fit can provide a meaningful local test while the generated spectrum and extraction still retain the full GP support.

The correlation calculation is a fixed-hyperparameter statement, not a claim that high-mass bins have mathematically zero influence. All retained bins still enter the global marginal likelihood and can shift the optimized length scale or kernel constant. A same-input diagnostic makes the aggregate effect explicit: after fixing the selected hyperparameters, removing the training bins above 150 MeV shifts the 65 MeV pull by a mean of -0.130 with a paired 90% interval $[-0.151, -0.110]$ in $1\% \times 10$, and by -0.195 $[-0.211, -0.179]$ in native 10%. The maximum absolute shifts are 0.376 and 0.400. In the first lane the mean changes from 0.139 to 0.009; in the second it changes from -0.246 to -0.441 , whose 90% interval $[-0.611, -0.271]$ crosses the adopted practical boundary. Counts above 150 MeV therefore have modest, statistically resolved collective influence even though every

pairwise RBF correlation is small. This diagnostic holds the chosen hyperparameters fixed and does not test their global-likelihood response to tail removal.

The optimized kernels nevertheless explain why a threshold-focused source fit is scientifically informative for the complete card: the low-mass extraction is directly coupled most strongly to the turn-on region, while the generated spectrum, specified smooth continuation, and full extraction retain the actual support. The restricted source fit thus provides a targeted conditional validation of local 65 MeV behavior under that full-card construction. It does not establish tail-independence or universal GP closure for every possible high-mass truth.

5.7 Matched refits and numerical selection

For every background, mass, and scheme, an independently optimized background-only fit supplies the matched reference uncertainty. Signals with $z = 1, 3, 5$ are generated with mean amplitude $A_{\text{inj}} = z\sigma_{A,\text{ref}}$, their full Gaussian templates are added to the same background counts, and the GP is refitted after applying the same training mask. The injected amplitude is therefore calibrated to the local background-only fit, whereas the pull denominator is always the uncertainty returned by the injected fit.

Whether a fit is kept never depends on its fitted signal or pull. Repeated attempts use the same pseudo-data with independent optimizer starts. We choose among them using the GP log marginal likelihood, reproducibility of the length scale and kernel constant, covariance validity, uncertainty-ratio checks, and contact with the kernel bounds. The choice never uses $\hat{A}$, the pull, agreement with the injected value, ε^2 , or a local p value. A statistical outlier is retained whenever it passes these numerical checks.

The historical study attempted 8000 fits and retained 7994; every cell contains 99 or 100 usable fits. The six rejected fits failed reproducibility checks. Each new continuation contains 400 usable fits—100 backgrounds at each of $z = 0, 1, 3, 5$ —with no rejection and no kernel-bound contact. Reaching those solutions required 614 optimizer attempts in the $1\% \times 10$ threshold-turn-on lane and 610 in the native 10% lane. Failed starts remain recorded, and checks of the covariance, counts, and pull calculation agree. Numerical selection is therefore independent of the physics result.

5.8 Full-100 spurious-signal results

Table 7 gives the two targeted 65 MeV results. The historical rows are shown only to explain why each replacement was required; no old and new pulls are combined. Both replacement samples exceed the 95-of-100 reporting threshold and pass the numerical checks above. We compare the results under the original statistical test, the post-result practical $|\bar{p}| < 0.5$ criterion, and the separate unit-width diagnostic.

The 100-background continuations sharpen, rather than simply confirm, the pilot interpretation. The $1\% \times 10$ threshold-turn-on truth is practically centered in the full sample and in the independent reserve-80, but its pull distribution is mildly overdispersed: the 90% width interval excludes one in both samples. Native 10% with 30–300 MeV support moves from the historical $+0.716$ response to -0.246 ; its full and reserve mean intervals both remain inside the practical band. The native full sample is statistically displaced under the targeted two-endpoint t/Holm screen, and the reserve-75 independently reproduces the negative direction. This is evidence for a small signed offset, not a failure of the adopted practical mean-closure margin. Moving the support and improving the turn-on description reduces the original 65 MeV offset enough to meet the post-result practical criterion, while exact unit pull-width calibration remains a separate caveat.

After substitution, all twenty displayed zero-signal point means satisfy $|\bar{p}| < 0.5$; the largest

Lane and 65 MeV scheme	n	$\bar{p}$ [90% CI]	s_p [90% CI]	mean assessment
1% $\times$ 10, historical	100	0.7890 [0.6293,0.9488]	0.9622 [0.8625,1.0907]	practical fail
1% $\times$ 10, threshold turn-on, full	100	0.1394 [−0.0499,0.3288]	1.1404 [1.0222,1.2927]	practical pass
1% $\times$ 10, threshold turn-on, reserve	80	0.0993 [−0.1150,0.3135]	1.1514 [1.0196,1.3265]	practical pass
native 10%, historical	100	0.7160 [0.5452,0.8867]	1.0284 [0.9218,1.1657]	practical fail
native 10%, 30–300 support, full	100	−0.2463 [−0.4165,−0.0761]	1.0253 [0.9190,1.1622]	practical pass [†]
native 10%, 30–300 support, reserve	75	−0.2840 [−0.4943,−0.0737]	1.0934 [0.9646,1.2661]	practical pass

Table 7: Targeted 65 MeV background-only closure. All headline intervals are two-sided 90% intervals: Student- t for the mean and exact normal-theory chi-square for the population standard deviation. Historical entries are unpaired context and are replaced, not pooled, in the composite validation figures. Both replacement full-100 mean intervals lie entirely within the post-result $[-0.5, 0.5]$ practical band, whereas the two historical means do not. [†]The native-10% full sample nevertheless has targeted Holm-2 $p = 0.03632$ and $|\bar{p}| > 0.2$; statistical resolution of a small offset and practical centering are reported as distinct statements. The 1% $\times$ 10 width interval excludes one in both the full and reserve samples.

absolute value is 0.338 at the historical 10% $\times$ 10, 180 MeV cell. Under the separate Holm-20 inferential screen that 180 MeV cell remains the only flag, while the native-10%, 65 MeV composite adjusted value is 0.291. The larger multiplicity family is not used to erase the targeted Holm-2 result. Rather, the note distinguishes practical magnitude from evidence that a nonzero offset is statistically resolved. The original two 65 MeV cells failed the practical margin with means 0.789 and 0.716; their replacements do not.

The same practical pattern holds when the correlated nonzero strengths are displayed: all 80 lane-mass-strength point means lie inside $[-0.5, 0.5]$, and 75 of their 80 pointwise 90% intervals are wholly contained. Exact unit-width calibration is less uniform. Three of the twenty zero-signal width intervals and fifteen of all eighty width intervals exclude one. At zero injection these are the replacement 1% $\times$ 10, 65 MeV cell, historical 1% $\times$ 10 at 90 MeV, and historical native 10% at 120 MeV. Thus practical mean/background-model-bias closure and exact normal-pull calibration are separate outcomes of the same figure.

Figure 31 directly compares the earlier and targeted 65 MeV ensembles, showing why the threshold-specific study was needed.

5.9 Recovery of injected signals

A centered zero-signal ensemble is necessary but not sufficient: a background model could remove the false offset and still absorb a real narrow signal. The nonzero rows therefore repeat the full GP fit after adding matched one-, three-, and five-reference-sigma signals. Figure 32 displays the mean and width at every strength and mass; Table 8 gives the numerical 65 MeV replacement results, including the paired response $R = (\hat{A}_z - \hat{A}_0)/A_{\text{inj}}$.

For the historical four-lane suite, the paired response separates signal recovery from the additive background offset. In the two replacements, the mean response is stable at 0.966–0.969

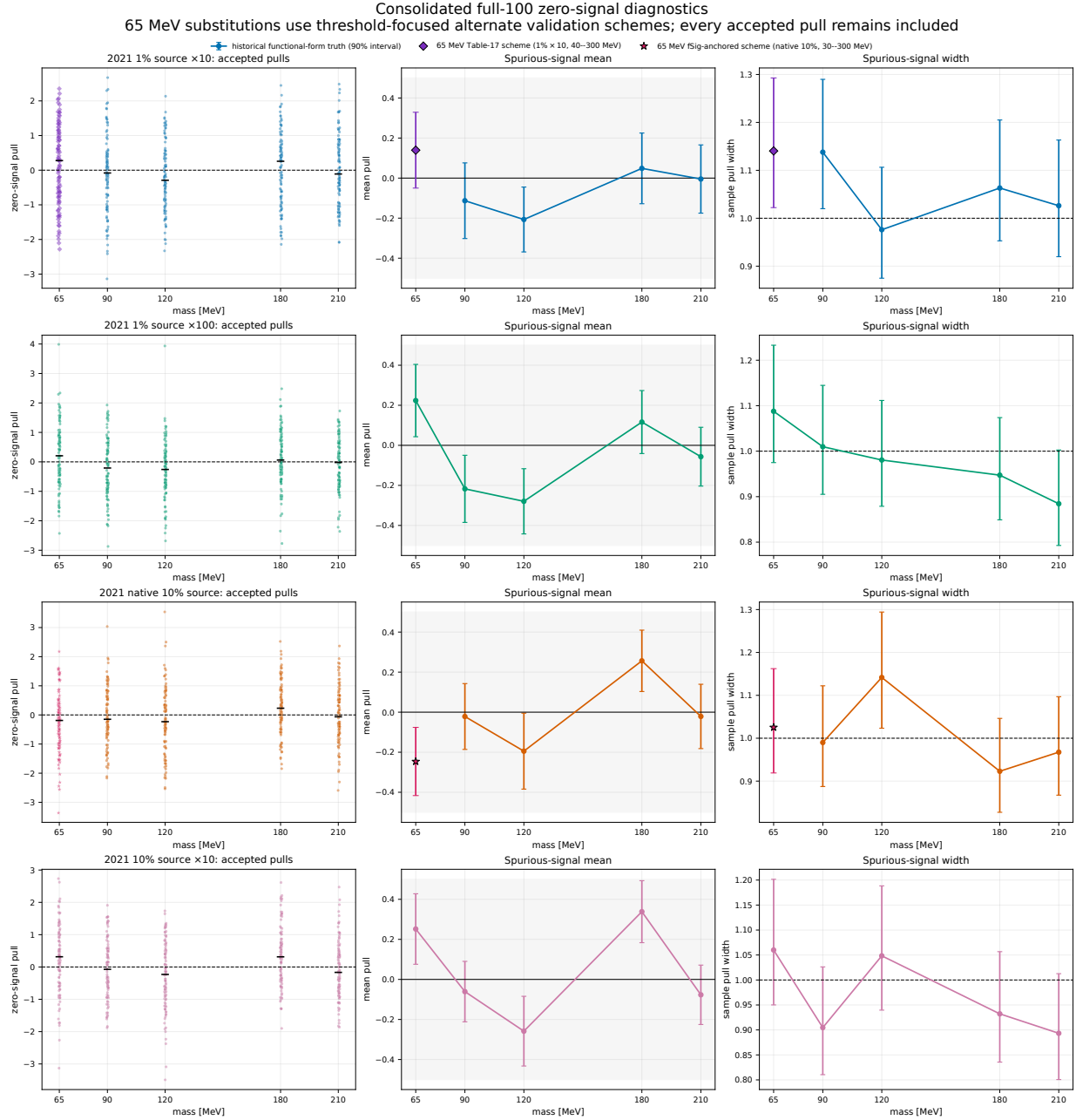


Figure 30: Full-100 zero-signal diagnostics for the four 2021 exposure lanes. The historical circular markers use the generalized-gamma threshold ensembles, fitted independently to the native 1% and native 10% sources. At 65 MeV only, distinct diamond and star markers replace—and are not pooled with—the historical values: 1% $\times$ 10 uses the full-100 dedicated threshold-turn-on continuation of the dedicated near-threshold study, while native 10% uses the full-100 threshold-focused extraction with 30–300 MeV GP support. The substituted points form a different, explicitly labeled validation scheme because the original analytic family did not adequately model the rising threshold. The panels retain all accepted pulls, two-sided 90% Student- t intervals for means, and exact normal-theory 90% chi-square intervals for widths; no accepted pull is clipped. The gray mean band is the post-result ± 0.5 practical centering tolerance. Connecting lines are visual guides only.

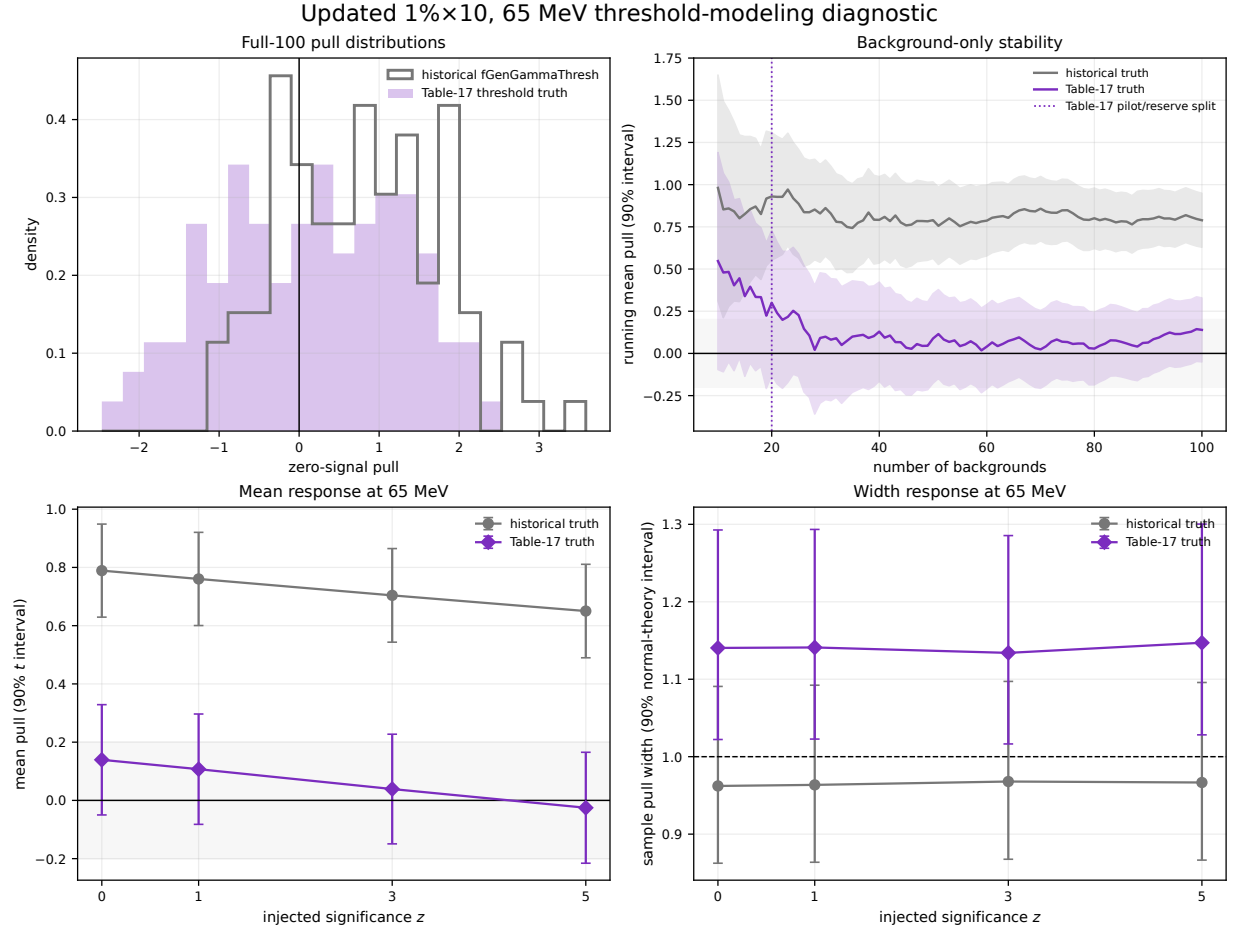


Figure 31: Targeted $1\% \times 10$, 65 MeV confirmation for the dedicated threshold-turn-on truth compared with the historical generalized-gamma threshold ensemble. The upper panels show the zero-injection pull distribution and the running mean; the lower panels show the mean and width across the correlated $z = 0, 1, 3, 5$ extractions. Mean intervals are two-sided 90% Student- t intervals and width intervals are exact normal-theory 90% chi-square intervals. The full-100 sample contains the original 20-background pilot; the reserve-only result in Table 7 demonstrates the same practical centering and the same mild overdispersion. Historical and replacement pulls are compared, not pooled. The gray ± 0.2 bands in this legacy-style focused display show the inherited material-effect reference; they are not the post-result ± 0.5 practical centering band used in this section. The native-10% replacement is shown in Figs. 30 and 32 and in Table 7.

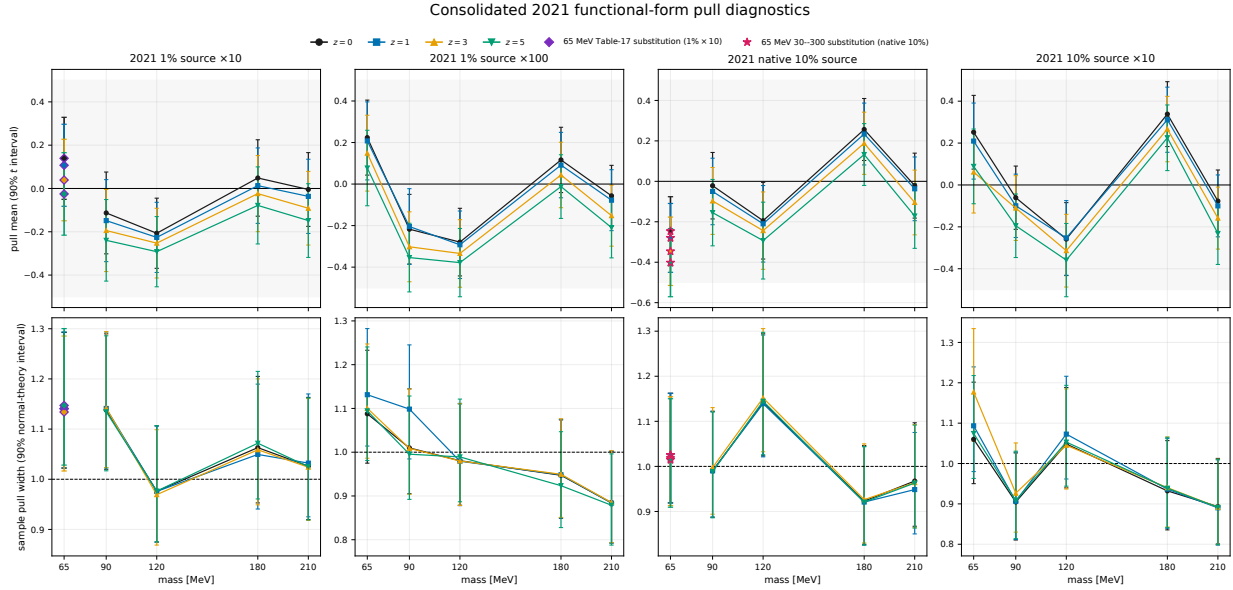


Figure 32: Compact composite background-validation summary. Columns correspond to $1\% \times 10$, $1\% \times 100$, native 10%, and $10\% \times 10$; the upper row shows pull means and the lower row pull widths as functions of mass. The four series are $z = 0, 1, 3, 5$, with zero injection visually emphasized and the correlated nonzero response series drawn more lightly. Circular markers are the original full-100 common-family generalized-gamma threshold ensembles. At 65 MeV, diamonds in $1\% \times 10$ and stars in native 10% replace the complete $z = 0, 1, 3, 5$ historical sets with the targeted studies defined in Table 6. Mean bars are two-sided 90% Student- t intervals and width bars are exact normal-theory two-sided 90% chi-square intervals. The substituted points are conditionally validated under different source-fitted truths; they are not a homogeneous pooled ensemble, and the four strengths share backgrounds. Connecting lines are omitted across scheme replacements. The gray mean bands show the post-result ± 0.5 practical centering tolerance.

over $z = 1, 3, 5$, with mutually consistent 90% intervals and no worsening with strength. The full procedure therefore recovers 96.6–96.9% of the tested amplitude, corresponding to a stable 3.1–3.4% attenuation. This is disclosed as a conditional response calibration, not rounded to exact unity. Agreement across strengths is not counted multiple times because the same 100 background realizations are reused.

65 MeV scheme	z	$\bar{p}$ [90% CI]	s_p [90% CI]	$\bar{R}$ [90% CI]
$1\% \times 10$, threshold turn-on	0	0.139 [−0.050,0.329]	1.140 [1.022,1.293]	—
	1	0.107 [−0.082,0.297]	1.141 [1.023,1.293]	0.9678 [0.9643,0.9713]
	3	0.039 [−0.149,0.227]	1.134 [1.016,1.285]	0.9665 [0.9646,0.9684]
	5	−0.025 [−0.216,0.165]	1.147 [1.028,1.300]	0.9670 [0.9654,0.9686]
native 10%, 30–300 support	0	−0.246 [−0.417,−0.076]	1.025 [0.919,1.162]	—
	1	−0.280 [−0.450,−0.109]	1.025 [0.919,1.162]	0.9664 [0.9629,0.9700]
	3	−0.346 [−0.515,−0.177]	1.019 [0.913,1.155]	0.9666 [0.9649,0.9684]
	5	−0.402 [−0.571,−0.234]	1.015 [0.909,1.150]	0.9686 [0.9669,0.9703]

Table 8: Full-100 65 MeV extraction at all matched-reference strengths for the two replacement schemes. Mean intervals are two-sided 90% Student- t intervals and width intervals are exact normal-theory two-sided 90% chi-square intervals. The response summaries are paired background-by-background and use two-sided 90% Student- t intervals. All strengths share the same backgrounds and are correlated. The $1\% \times 10$ widths are mildly high at every strength; the native-10% widths are compatible with one at this precision.

5.10 What the validation establishes

Taken together, these tests show practical mean closure and stable signal recovery for the source-like spectra that were studied. The training exclusion keeps the GP from learning the signal tails, the pseudo-data means are fitted independently of the GP, and every pseudo-spectrum receives a fresh sideband fit. Numerical branches are chosen without reference to the fitted signal or pull. Near threshold, the historical analytic family is replaced only in the two cells where its description of the source is inadequate; tables and figures identify those replacements explicitly.

Under the adopted practical criterion, the near-threshold concern is resolved within this source-conditioned suite. The two replacement 100-background intervals and their independent reserve samples lie inside the ± 0.5 mean-pull band. So do all twenty zero-signal point estimates and all eighty correlated injected-strength point estimates in the four-lane display. The paired response remains 96.6–96.9%. The practical mean-bias requirement is therefore met for these tests, provided that each replacement remains identified by its generating model and GP support.

The stricter diagnostics remain important. The original $10\% \times 10$, 180 MeV cell retains an exploratory Holm-20 flag with mean 0.338, and the native-10%, 65 MeV replacement retains a targeted Holm-2 flag. Three of twenty zero-signal width intervals and fifteen of eighty all-strength width intervals exclude one; the stable signal attenuation is 3.1–3.4%; and the fixed-hyperparameter tail-removal shifts are nonzero. The study therefore does not prove zero bias for arbitrary smooth spectra, exact pull calibration, confidence-limit coverage, bias in the observed data, or a scan-global significance calibration. Any larger study should retain the source, support, optimizer, and post-result tolerance labels used here.

Validation component	Evidence	Interpretation
Training leakage and blind geometry	train-2.25, widened extraction window, recovery and leakage scans	supports the production mask
Functional-form source description	source-fit and bin-integral diagnostics; threshold-specific replacements at 65 MeV	the 1%-source threshold model is data-like over the support; the native-10% model is locally data-like and conditional on its tail
Spurious-signal means	Fig. 30 and Table 7	all 20 zero-signal point means meet $ \bar{p} < 0.5$; both replacement 90% intervals lie inside the band
Pull widths	exact normal-theory 90% chi-square intervals in Fig. 32	exact unit-width calibration is not global: 3/20 zero-signal and 15/80 all-strength intervals exclude one
Signal injection	paired $z = 1, 3, 5$ response with a fresh GP fit	stable 96.6–96.9% response; tests extraction, not direct confidence-limit coverage
Numerical stability	multistart and reproducibility checks independent of the fitted signal	all 800 fits in the new lanes are usable
High-mass influence on threshold	direct RBF correlations and fixed-hyperparameter tail-removal diagnostic	local validation is meaningful, but the aggregate tail effect is nonzero and tail independence is not claimed
Observed inference	no observed spectrum or limit enters truth/branch selection	unchanged by these conditional toys

Table 9: Summary of the background-validation evidence. The practical mean criterion, statistical tests, pull-width calibration, signal response, and tail influence are kept separate; none is promoted into a coverage or universality claim.

5.11 Selecting the 2021 GP support

The signal search tests masses from 50 to 250 MeV, but the GP also needs data outside that interval to constrain the background near its endpoints. At low mass, the native 2021 spectrum has a sparse shoulder followed by a rapid turn-on: its 0.125 MeV bins contain only a few hundred events around 30–32 MeV and several thousand by 40 MeV. We therefore compared several lower support edges while holding the upper edge at 300 MeV and leaving the search interval unchanged. The aim was to exclude the least stable part of the turn-on without discarding useful sideband information.

5.11.1 Source shape and candidate supports

Figure 33 shows the source spectrum and the tested lower edges. The 30 MeV setting is retained as a control but was not eligible for selection; the candidate edges were 32, 34, 36, 38, and 40 MeV.

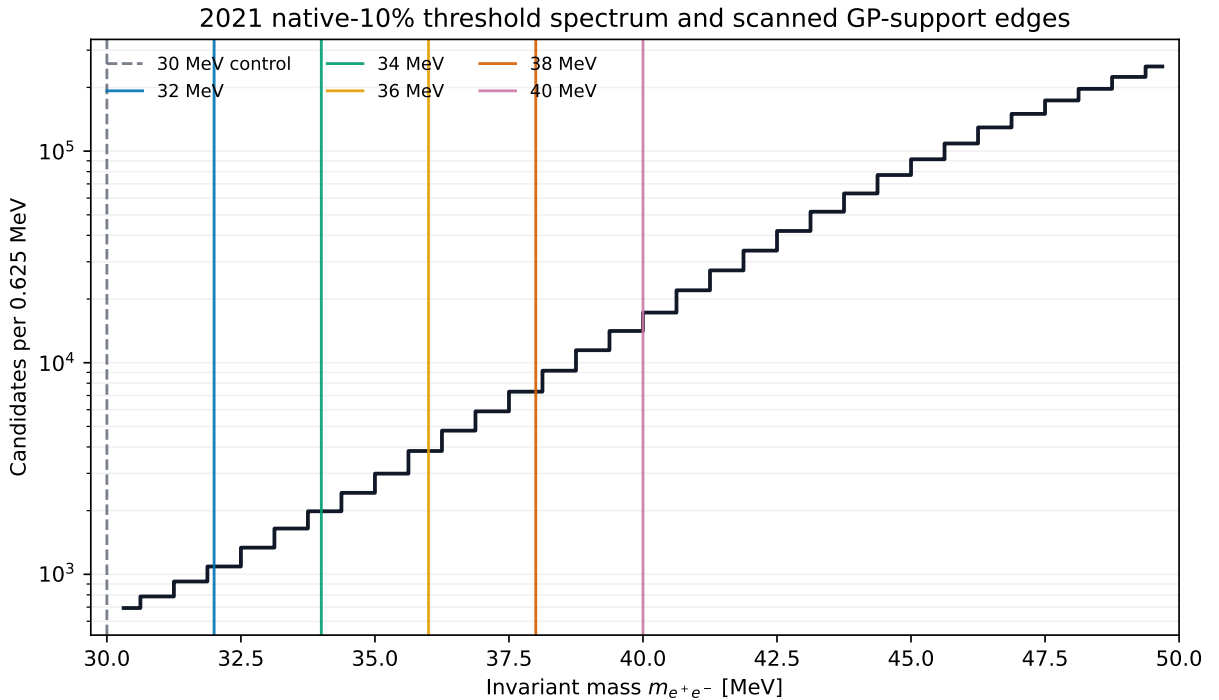


Figure 33: Native 2021 10% source histogram near threshold. Vertical lines mark the 30 MeV control and the five candidate lower edges of the GP support. The logarithmic vertical scale makes the low-count shoulder and rapid turn-on visible.

5.11.2 Initial scan

The first stage reuses the 100 source-conditioned backgrounds archived with v4.9.1; no new pseudo-data are generated. At each support edge, matched-reference extractions are performed at 55, 60, 65, and 70 MeV with 0-, 2-, and 5- σ_A injections. The scan changes only the lower support edge. Its binning, preprocessing, blind and training masks, kernel bounds, signed extraction, and optimizer checks are inherited from the v4.2/v4.5 analysis definition. Optimizer branches are chosen using fit reproducibility, covariance validity, kernel state, and GP log marginal likelihood; neither the pull nor any limit or p -value enters that choice.

The first 25 backgrounds were processed at all six edges. All 1800 fits passed the numerical checks, with no invalid covariance matrix or kernel-bound contact. Under the original rule, an edge was rejected when any injection cell had $|\bar{p}| \geq 0.5$ and its two-sided 90% Student- t interval excluded zero. Every candidate was rejected. Table 10 records that outcome under the rule used for the initial scan.

Table 10: Initial 25-background minimax summary. Every candidate is rejected by the original injection criterion. The 30 MeV row is a control and was never eligible for selection.

Lower edge [MeV]	Worst $ \bar{p} $	Worst zero-signal $ \bar{p} $	Initial rule
30	1.624	1.300	fail (control)
32	1.530	1.331	fail
34	1.447	1.277	fail
36	1.134	0.959	fail
38	1.072	0.864	fail
40	1.152	1.152	fail

5.11.3 Practical criterion and independent confirmation

After the first stage, we adopted a less stringent practical criterion: at least 9 of the 12 mass-injection means and at least 3 of the 4 zero-signal means had to satisfy $|\bar{p}| < 0.75$, and no mean could reach 1.25 in magnitude. The numerical checks, the minimax score, and the 0.10 tie margin were left unchanged. Among candidates within that margin, the lower edge was preferred so that the fit retained more data. Because this criterion was formulated after the first 25 backgrounds had been examined, it is a post-selection practical rule, not a predeclared equivalence test or a coverage criterion. It was fixed before the remaining 75 backgrounds were processed.

Only the 36 and 38 MeV candidates pass the practical rule in the first stage. The 38 MeV edge has the smaller worst-cell pull, but 36 MeV lies within the retained tie margin and is therefore selected for confirmation. The independent continuation uses backgrounds 25–99 at 34, 36, and 38 MeV. At 36 MeV, 10 of 12 cell means and all four zero-signal means satisfy the 0.75 condition in the continuation alone; the worst magnitude is 0.901. In the complete 100-background sample, the corresponding counts are 9 of 12 and 3 of 4, with a worst magnitude of 0.960. Every fit passes the numerical checks. With the criterion already fixed, these results freeze the 2021 GP support at 36–300 MeV without further tuning.

At the selected edge, the full-sample means at 55 MeV are -0.773 , -0.841 , and -0.960 for the 0-, 2-, and $5\text{-}\sigma_A$ injections. Their 90% intervals exclude zero, so the low-mass under-recovery is retained as a real limitation of this test. Across the nine cells from 60 to 70 MeV, the means range from -0.110 to $+0.325$ and the pull widths from 0.939 to 1.034. The 65 MeV zero-signal mean is 0.253, with a 90% interval of $[0.086, 0.421]$.

Each accepted toy also provides a bounded Wald $\tilde{q}_\mu$ 90% CL_s yield limit. We use those values only to check that the limit is stable under nearby support choices; the smallest limit never determines the selected edge. This study therefore supports the 36–300 MeV choice for the declared source-conditioned ensemble, with the 55 MeV caveat above. It does not establish direct coverage or truth-independent background closure.

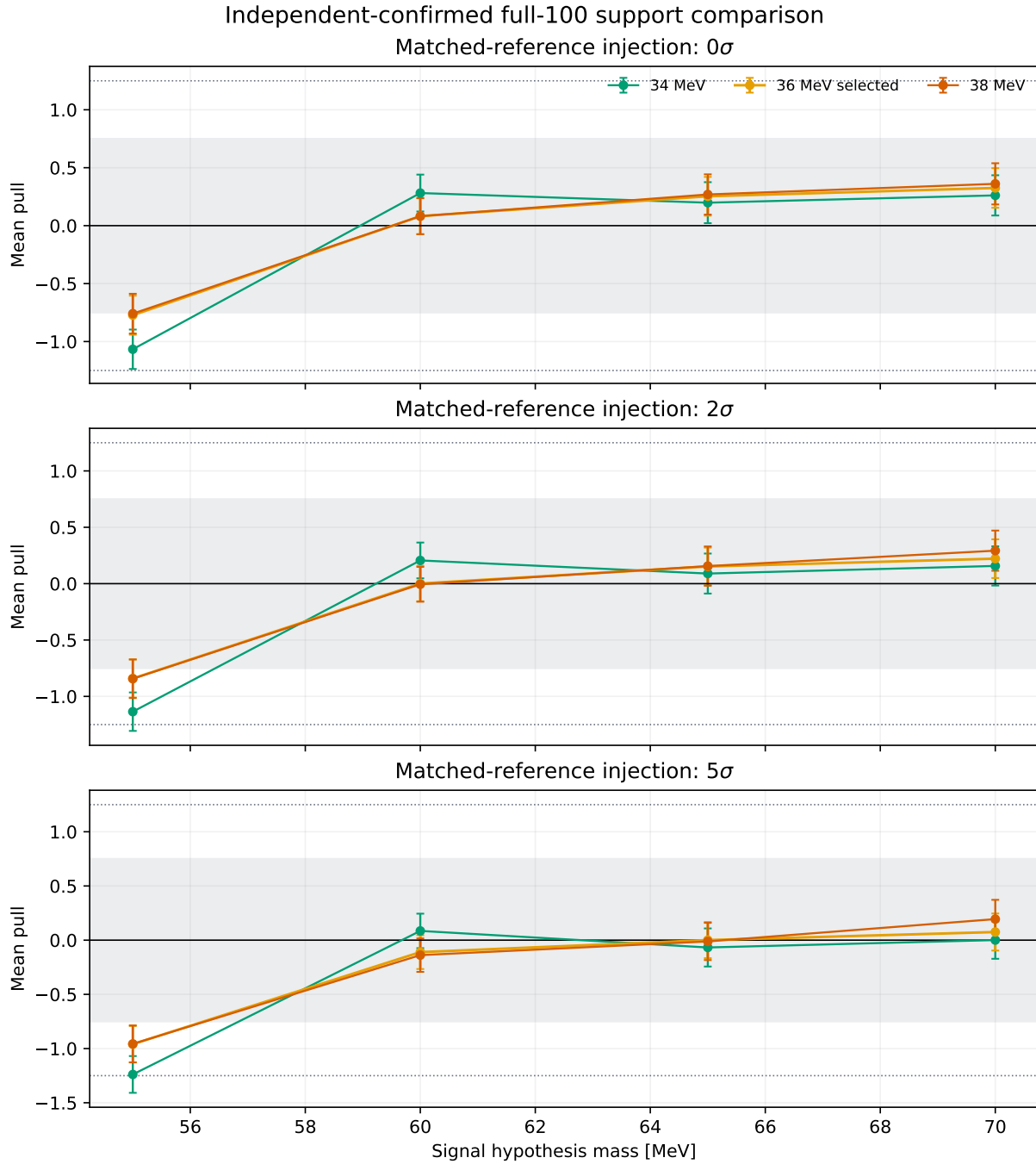


Figure 34: Mean pulls in the complete 100-background sample for the selected edge and its two neighboring candidates. Error bars are two-sided 90% Student- t intervals on the mean. The shaded band denotes the post-selection $|\bar{p}| < 0.75$ practical region; dotted lines show the magnitude-1.25 guard. The remaining exceptions are confined to the 55 MeV hypothesis.

6 Selected observed results

This section brings the principal observed results into one compact comparison. At each mass hypothesis, the upper curves answer two versions of the same question: how many signal events remain compatible with the data under the 90% CL_s criterion, and what upper limit does that imply for the kinetic-mixing parameter ε^2 ? The lower curves show the corresponding local asymptotic background-only p_0 . A small p_0 identifies an upward fluctuation worth examining; it is not, on its own, a discovery probability for a scan over many masses. Only observed curves are shown. There are no expected-limit bands in any figure in this section.

The curves deliberately preserve two distinct analysis states. The full-2015 and full-2016 results, and every multi-campaign combination, use the reviewed historical v4.2 state. In that state, the 2021 contribution is the native 10% sample with a 40–300 MeV Gaussian-process (GP) support. The separate 2021 10% individual curve uses the later v4.9.5 result with the selected 36–300 MeV support. It is displayed as the most recent standalone 2021 result, but it is not substituted into the historical combinations. The 2016 10% and 2021 1% curves are likewise standalone comparisons. A later source-conditioned attempt to choose a new support for the full-2016 statistics did not satisfy its acceptance rule, so it provides no updated full-2016 observed result. Retaining the reviewed v4.2 curve avoids turning that failed optimization into an unqualified analysis choice.

6.1 Individual data sets

Figure 35 compares five standalone scans: full 2015, 2016 10%, full 2016, 2021 1%, and 2021 10%. The upper panel reports the signal-yield limit before translating it into a coupling; the middle panel applies the campaign-specific luminosity, efficiency, resolution, radiative fraction, and branching inputs needed for the ε^2 limit. Keeping both panels visible separates what comes directly from the statistical fit from what also depends on the physics normalization.

The lowest three mass hypotheses in the 2021 1% scan, 50–52 MeV, lie immediately above its lower search boundary and are sensitive to the treatment of the GP support edge. They are therefore shown for completeness but not promoted as candidate-level evidence. This distinction matters especially in a multi-curve plot: a formally small local value at an analysis boundary can be visually striking even when it is known to be a modeling diagnostic.

After those three edge points are set aside, the smallest local value in the 2021 1% series occurs at 244 MeV, $p_0 = 2.616405 \times 10^{-3}$ ($Z = 2.79234$). In the current 2021 10% series, the minimum occurs at 78 MeV, $p_0 = 2.474845 \times 10^{-3}$ ($Z = 2.81029$). Both are fixed-mass summaries rather than scan-wide significances.

6.2 Shared-coupling combinations

The combinations in Figure 36 profile one common ε^2 while allowing each campaign to retain its own expected signal yield and background model. They are simultaneous likelihood results, not pointwise envelopes or arithmetic averages of the standalone limits. The three pairwise curves are full 2015 + 2021 10%, full 2016 + 2021 10%, and full 2015 + full 2016. The all-three curve is full 2015 + full 2016 + 2021 10%, evaluated where all three campaigns overlap. It does not imply a 2021 full-statistics result.

Using a common coupling is important physically. A candidate signal would have the same ε^2 in each campaign, but its expected event count would change with beam conditions, exposure, acceptance, and mass resolution. The simultaneous likelihood enforces that relationship while allowing each data set to contribute according to its own reach.

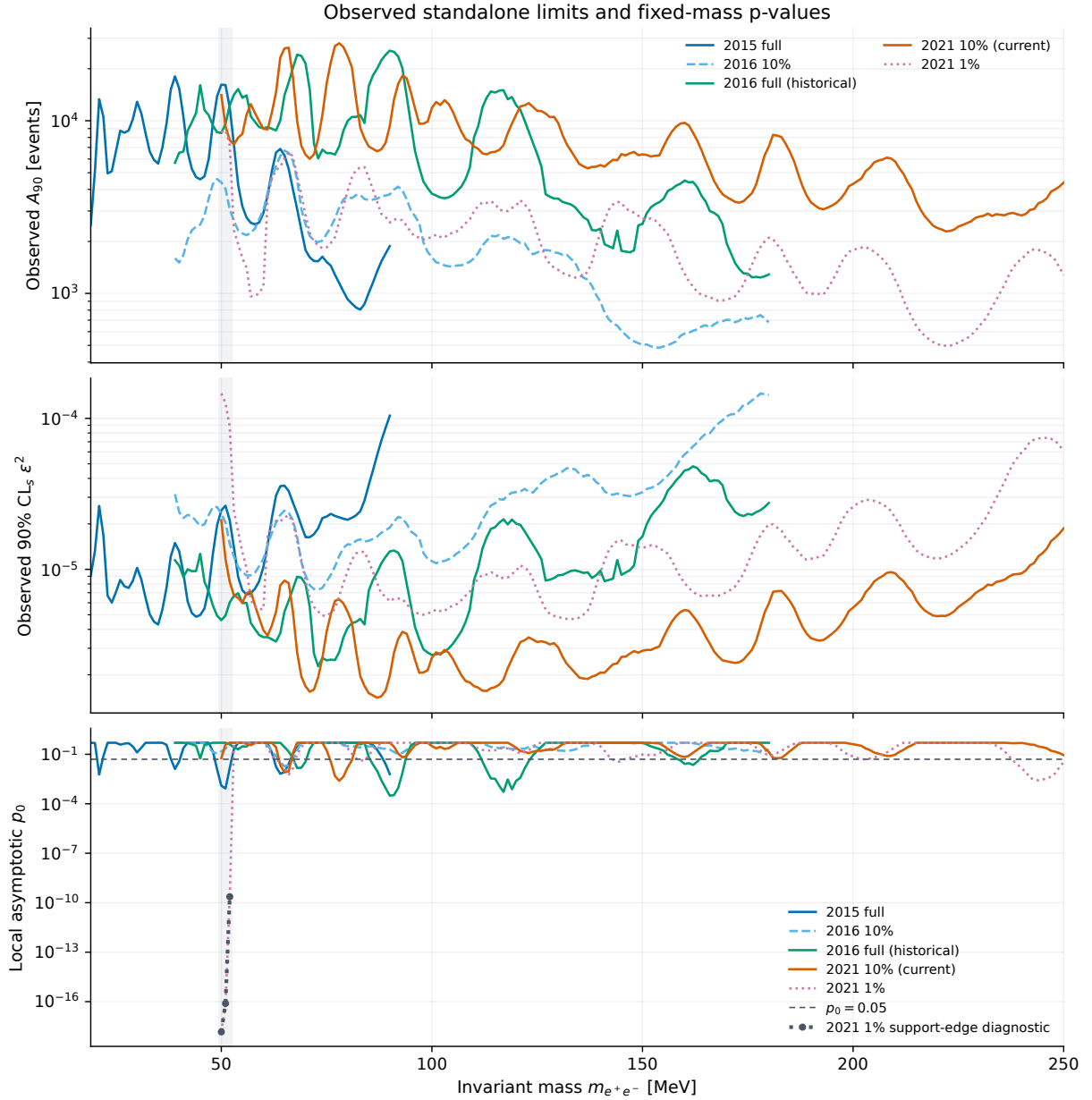


Figure 35: Standalone observed results for full 2015, 2016 10%, full 2016, 2021 1%, and 2021 10%. From top to bottom, the panels show the 90% asymptotic CL_s signal-yield upper limit, the corresponding electron-channel ϵ^2 upper limit, and the local asymptotic p_0 . The full-2015 and full-2016 curves are the historical v4.2 results; the 2021 10% curve is the later v4.9.5 standalone result. No expected-limit bands are shown. The marked 50–52 MeV portion of the 2021 1% scan is retained as a GP-support-edge diagnostic and is excluded from candidate interpretation.

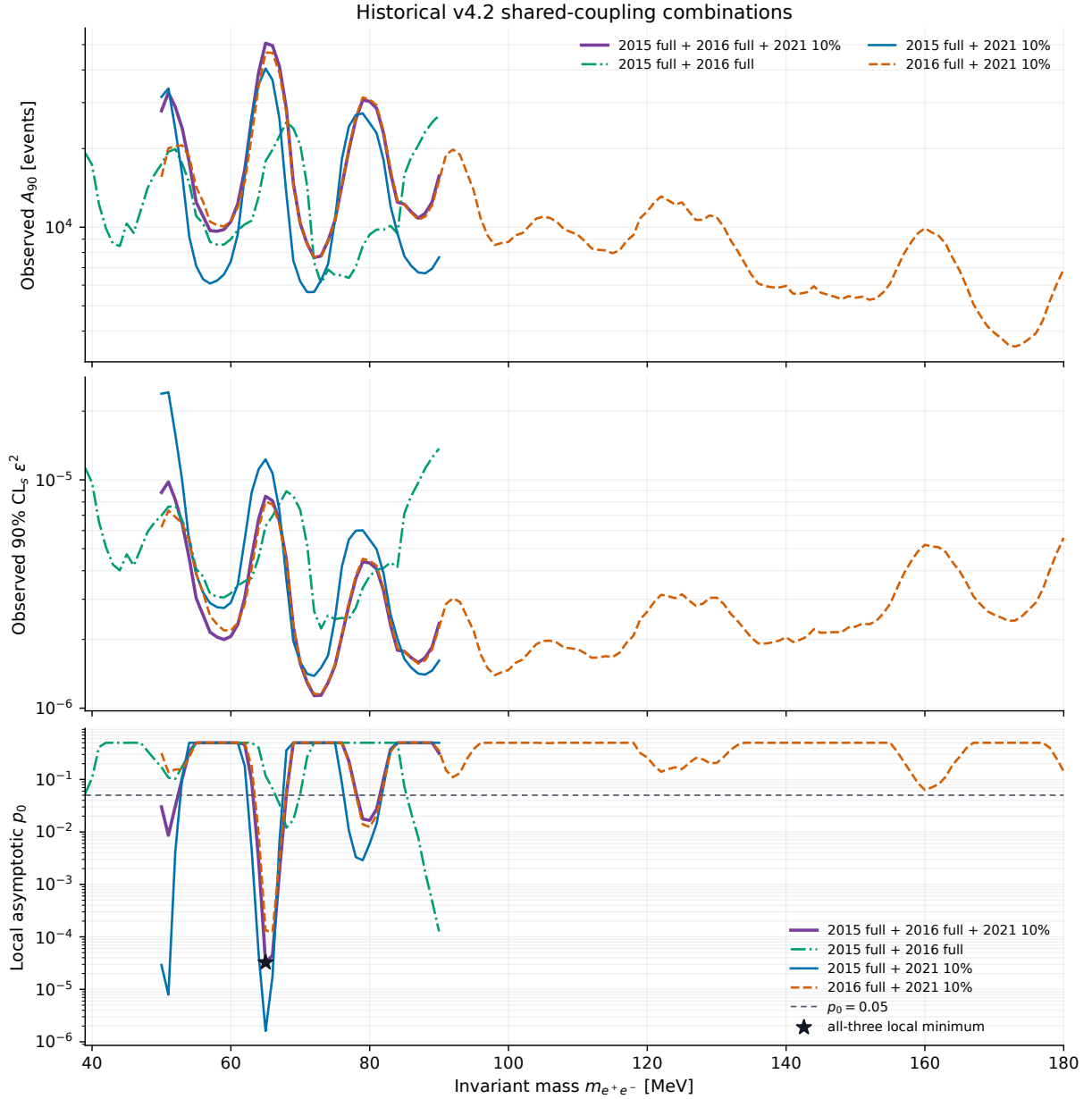


Figure 36: Historical v4.2 shared- ε^2 results for the three pairwise combinations and the common three-campaign overlap. For visual alignment, the top coordinate multiplies the shared- ε^2 limit by the sum of the active campaigns' signal normalizations. The remaining panels show the observed 90% asymptotic CL_s upper limit on ε^2 , and the local asymptotic p_0 . Each combination uses the reviewed v4.2 campaign states, including the 40–300 MeV GP support for the 2021 10% contribution. No expected-limit bands are shown.

6.3 Local asymptotic scan

Figure 37 collects the local asymptotic p_0 series without the two limit panels. This view makes common mass-local structure easier to compare across data sets and combinations. Every point tests one fixed mass. The series has not been calibrated with scan-wide background-only pseudoexperiments, and no look-elsewhere correction is assigned here. Consequently, the plotted values are local diagnostics rather than global significances. They also should not be read as coverage tests for the upper limits.

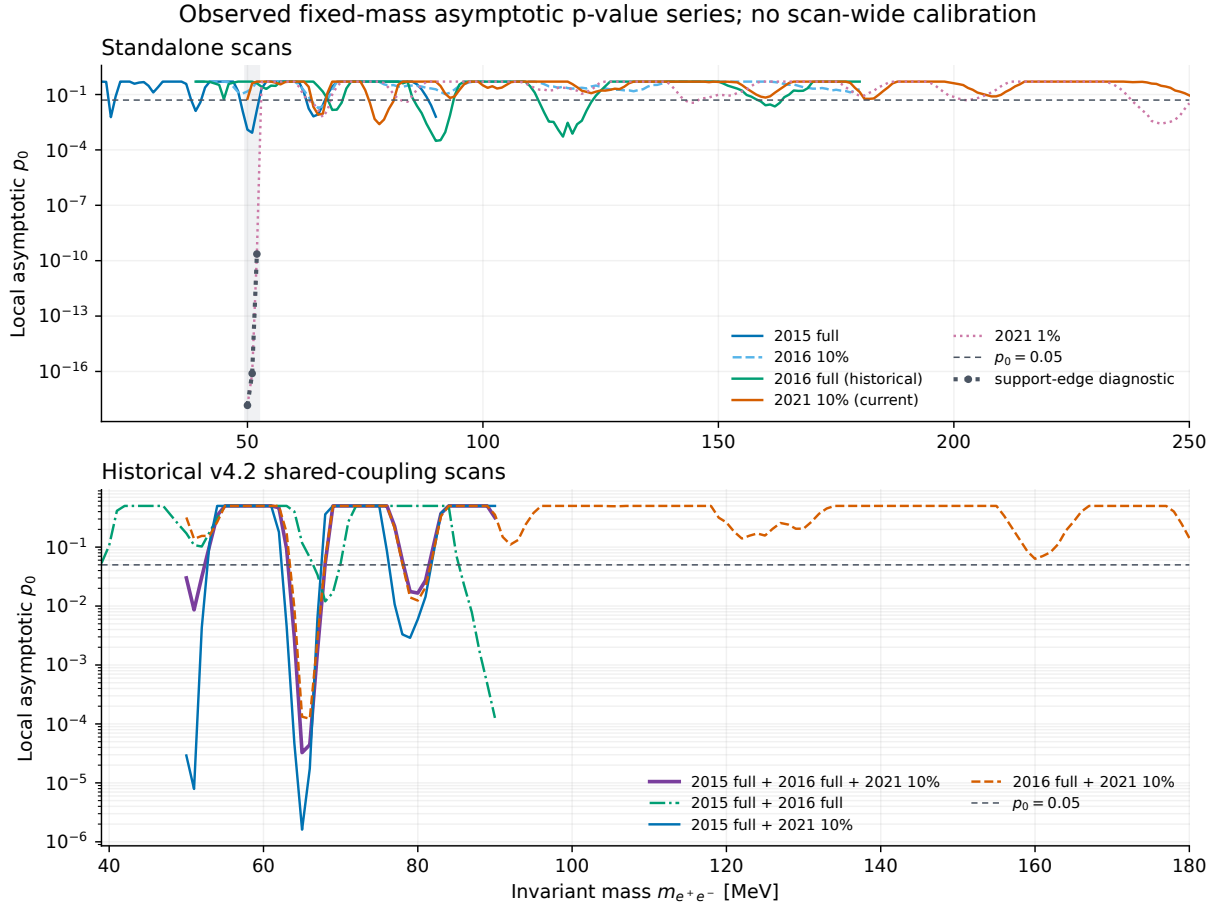


Figure 37: Local asymptotic background-only p_0 as a function of mass for the standalone scans and the historical v4.2 shared-coupling combinations. The support-sensitive 50–52 MeV segment of the 2021 1% scan is marked as a boundary diagnostic. These are fixed-mass values: no scan-wide calibration or global significance is reported.

6.4 The largest all-three local fluctuation

Within the historical v4.2 all-three overlap, the smallest local asymptotic value occurs at 65 MeV: $p_0 = 3.259183 \times 10^{-5}$, corresponding to $Z = 3.99321$ under the one-sided local asymptotic mapping. Figure 38 shows the underlying extraction separately in the three campaigns. Presenting the data, GP background, and fitted signal contribution together makes it possible to see how a single shared coupling is supported—or constrained—by statistically independent spectra.

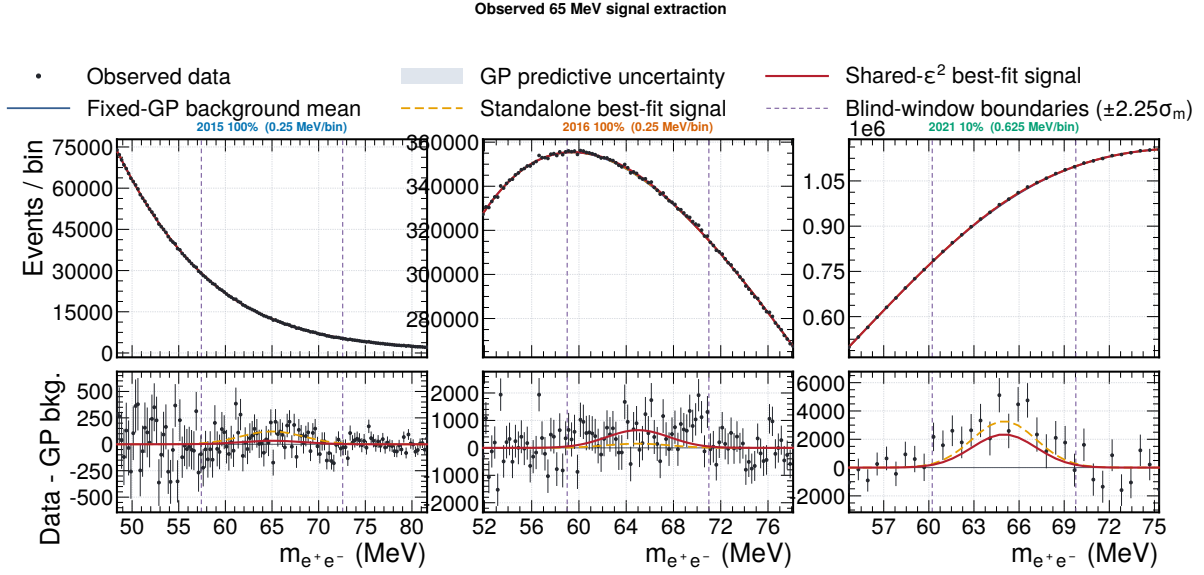


Figure 38: Historical v4.2 extraction at the 65 MeV mass hypothesis, the largest local upward fluctuation in the all-three overlap. The three panels show the full 2015, full 2016, and 2021 10% spectra with their GP background descriptions and the signal contributions associated with standalone and shared- ε^2 fits. The quoted p_0 and Z are local asymptotic quantities. This display is a candidate diagnostic, not a scan-calibrated discovery claim.

Taken together, these plots show both the gain from combining campaigns and the discipline required to interpret that gain. Upper limits may be compared only within their stated analysis configurations, while localized excesses require an additional scan-wide calibration before they can support a global claim.

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